

# A Conditional Sixth-Order Lie-Group FullVA Integrator for Lower-Pair Mechanisms

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## Abstract

Lower-pair mechanisms require Lie-group integrators that keep position, velocity, and acceleration constraints synchronized. This paper studies Gauss6/FullVA, a three-stage Gauss collocation integrator whose stage system couples absolute-coordinate stage kinematics, Newton–Euler balance, and lower-pair constraints at position, velocity, and acceleration level in one square nonlinear system with Lie-group endpoint reconstruction. We show that the implemented stage system is exactly three-stage Gauss collocation of the reduced joint-coordinate equation of motion written in absolute coordinates: the lifted reduced Gauss stage satisfies all 132 stage rows identically. Building on this identity, the main theorem proves a conditional sixth-order same-initial-state reported-grid estimate for the selected smooth branch. The local defect is the sum of the classical Gauss endpoint defect, an endpoint-closure perturbation, and an inexact-Newton perturbation; the retained hypotheses are compact-tube regularity, uniform local inverses, a one-step stability scale, and the branch-selected solver envelope  $\eta_h \leq c_\eta h^7$ . The algebraic stage identity, the perturbation lemmas with their explicit constants, the Gauss tableau order conditions, and the quadrature-defect bound are machine-checked in Lean 4/Mathlib. On a smooth cylindrical-chain benchmark the implementation records position/velocity orders 7.161/7.066 as branch-consistency diagnostics. Four ASME-style mechanisms exercise the same residual vocabulary: single- and double-pendulum rows provide dynamic-order evidence, while four-link and slider-crank rows provide closed-loop constraint/reaction consistency. Residual-to-error promotion, full external same-test reproduc-

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tion, and complete source-paper TFE residual replacement are not claimed; the comparison to TFE is kept at the formal-order and diagnostic levels.

*Keywords:* Lie group integrator, index-3 DAE, multibody dynamics, lower-pair mechanisms, Gauss collocation, temporal finite elements

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## 1. Introduction

Implicit integration of constrained multibody systems must keep position, velocity, and acceleration constraints synchronized while respecting the geometry of finite rotations. This is especially important for absolute- or lower-pair formulations, where a step update that looks accurate in position can still carry endpoint velocity drift or inconsistent reaction dynamics. Recent time finite element methods on Lie groups have shown that high-order integration of frictional index-3 differential-algebraic equations is feasible and practically valuable [1]. The present work asks a more specific numerical-methods question: can a Gauss-collocation Lie-group FullVA residual be organized so that lower-pair position, velocity, and acceleration constraints are enforced monolithically while retaining a theorem-level conditional sixth-order claim on the retained smooth branch?

The resulting method is Gauss6/FullVA, a Gauss collocation FullVA integrator whose sixth-order claim is tied to the branch smoothness, endpoint/stability, and nonlinear-solver contracts stated below. Its novelty is not the existence of Gauss collocation itself; it is the monolithic Lie-group FullVA residual organization for lower-pair mechanisms, where Gauss stage kinematics,  $\text{SO}(3)$  retraction variables, Newton–Euler dynamics, and position/velocity/acceleration lower-pair rows are solved in one square nonlinear system. Temporal finite elements enter this paper as a formal-order comparator only after the method and theorem have been fixed. Their role is to mark the local  $m = 3$  order-five formula target and the diagnostic boundary for separate same-test reproduction, not to supply an implementation superiority claim, a full-TFE replacement, or a theorem premise. Accordingly, the central mathematical object is the branch-selected Gauss6/FullVA residual together with its retained-interface order theorem. The comparison and reproducibility material states the formal-order, external same-test, and source-package scope without modifying the theorem. The proof organization follows the Lie-group constrained-BDF literature [8] only as an obligation ordering: fix the discrete object, insert the smooth branch, prove

the local defect, and then propagate it globally. A structural fact makes the first two steps short here: the implemented stage system is exactly three-stage Gauss collocation of the reduced joint-coordinate equation of motion written in absolute coordinates, so the lifted reduced Gauss stage satisfies every stage row identically.

The contributions are as follows.

1. a monolithic square 132-row Lie-group FullVA stage residual for lower-pair mechanisms, together with the exact stage identity showing that its stage solve is reduced joint-coordinate Gauss collocation in absolute coordinates;
2. a complete one-step algorithm and conditional sixth-order same-initial-state reported-grid theorem obtained by local-defect-to-global-error transfer on the compact-tube Gauss-predictor branch under retained endpoint/stability and solver-scale interfaces, with the algebraic identities, perturbation lemmas, tableau order conditions, and quadrature bound machine-checked in Lean 4/Mathlib, and with finite-window smooth slopes 7.161/7.066 reported only as branch-consistency diagnostics;
3. method-side dynamic-order evidence for the single- and double-pendulum examples, and mechanism-coverage evidence for the four-link and slider-crank examples under the same FullVA vocabulary;
4. a scope boundary that limits TFE material to the local order-five  $m = 3$  formula target and keeps implemented source-paper comparisons diagnostic until the same-test promotion criteria are met.

A separate reproducibility appendix records external reproduction, full-TFE, and source-package boundaries. It is included to delimit non-claims and reproduce the accepted rows, not as an additional method contribution or theorem input.

Table 1 fixes the hierarchy used throughout the paper. The proof contribution is the analytic residual-to-local-defect transfer for the implemented Gauss6/FullVA map. Numerical rows are downstream evidence for the accepted path. External same-test reproduction, source-package, and full-TFE replacement records are boundary evidence; they keep stronger external claims out of the manuscript and are not premises of the theorem. The load-bearing proof is the direct local-defect-to-global-error argument for that same-initial-state reported-grid theorem.

**Table 1: Claim hierarchy used in the manuscript.**

| Evidence layer                                   | Role in the paper                                                                                                                                                                           | Claim role                                                                            |
|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Gauss6/FullVA residual and theorem               | Defines the 132-row Lie-group FullVA stage solve and the conditional sixth-order same-initial-state reported-grid proof obtained from compact-branch local-defect-to-global-error transfer. | Main contribution.                                                                    |
| Smooth cylindrical-chain and pendulum rows       | Check the accepted smooth path and provide method-side dynamic-order evidence for the examples whose trajectory-order policy is accepted.                                                   | Primary numerical evidence.                                                           |
| Four-link and slider-crank rows                  | Exercise closed-loop lower-pair constraints, reaction consistency, and local coarse dynamics under the same residual vocabulary.                                                            | Mechanism coverage, not standalone asymptotic order proof.                            |
| TFE and common-reference comparisons             | Place the method against the local $m = 3$ order-five formula target and record bounded diagnostic comparisons.                                                                             | Formal-order and diagnostic comparison only.                                          |
| External reproduction and source-package records | Record why external same-test superiority, full-TFE replacement, and residual-to-error promotion are outside the accepted theorem and numerical claim.                                      | Claim-scope and reproducibility boundary, not a method contribution or theorem input. |

Throughout the paper, “conditional order” has this narrow meaning: a local-defect and local-to-global consistency transfer for the accepted map under the stated compact-tube, endpoint-closure, and branch-solver assumptions. It is an analytic perturbation statement, not an empirical order fit: the smooth trajectory slopes are reported only after the theorem-level estimates have been stated. It is not a residual-to-error transfer theorem for mechanism-coverage rows, not a proof that fixed production tolerances are asymptotically sufficient, and not a full source-paper same-test TFE repro-

duction. The P7 residual-to-error boundary is therefore a downstream output nonclaim, not one of the retained branch-domain hypotheses used to prove the smooth-branch theorem. The one-step map is defined only for transitions that remain on the selected Gauss-predictor branch and satisfy the compact-tube, endpoint-closure, and scaled Newton residual conditions in the theorem. Transitions outside that domain are not assigned a theorem-level order by residual tables, fixed tolerances, or external comparison rows. The scaled Newton condition is a proof-norm residual condition imposed before endpoint or trajectory errors are evaluated; it is not an assumed local-defect or grid-error conclusion.

**Proof structure.** The proof has three steps. First, Lemma 1 shows that the implemented stage system is exactly three-stage Gauss collocation of the reduced joint-coordinate equation of motion, written in absolute coordinates: the lifted reduced Gauss stage satisfies all 132 stage rows identically, so no stage-residual perturbation term appears. Second, the two implementation-side perturbations, endpoint closure and inexact Newton, are bounded by  $O(h^7)$  under explicit local hypotheses. Third, the classical Gauss endpoint defect and these two perturbations are transferred to the reported grid by a discrete Gronwall argument with tube retention. The retained interfaces are stated once, in Assumption 1, and fixed before any numerical evidence is read. Two of them are not independent: P2 follows from P1 for small steps (Lemma 3), and the solver envelope P6 is reached by the Newton iteration from any predictor in the contraction ball under P1 and P2 (Lemma 7); both are kept as stated interfaces because the theorem is formulated for any mechanism with this stage structure, because the implemented predictor’s ball membership is checked only a posteriori, and because the production runs use a fixed tolerance. The algebraic identities and the perturbation lemmas are machine-checked in a Lean 4/Mathlib development distributed with the source package; Butcher’s order theorem and the classical Gauss local-error estimate enter as cited results.

The remainder of the paper follows the evidence chain in that order. Section 5 defines the accepted one-step map and the complete execution algorithm. Section 6 fixes the validation policy, including the rule that external comparisons remain diagnostic unless the same-test promotion criteria are satisfied. Section Appendix A records the detailed same-test promotion boundary. Section 7 proves the exact stage identity and the conditional sixth-order theorem. Sections 8 and 9 report the numerical evidence. The final diagnostic and reproducibility sections document what has been checked

without turning those checks into broader claims.

## 2. Related work

The method sits at the intersection of geometric integration, constrained multibody dynamics, collocation methods for differential-algebraic equations (DAEs), variational integration, nonsmooth contact dynamics, and time finite elements. The relevant literature separates into six clusters, each of which leaves a different gap.

First, Lie-group integrators preserve the rotation manifold by updating in a local tangent chart rather than advancing redundant orientation coordinates and then repairing them. Runge–Kutta–Munthe-Kaas methods provide a general high-order construction on manifolds [5, 6]. Multibody-specific work has adapted this geometric viewpoint to constrained mechanics through Lie-group generalized-alpha schemes [7], BDF-type schemes on Lie groups [8], and quaternion-based singularity-free rotational integration [9, 10]. These methods establish how to evolve rotations geometrically, but they do not by themselves prescribe a stage residual that simultaneously enforces lower-pair position, velocity, and acceleration consistency in one Newton solve.

Second, constrained multibody systems in absolute or redundant coordinates are index-3 DAEs. Their numerical order is affected not only by the nominal order of the time integrator, but also by how the algebraic constraints and their differentiated forms are enforced [3]. The absolute-coordinate Lie-group formulations of Kissel, Taves, Fang, Zhang, and Negrut clarify the connection between small body-frame rotation increments and the first-order sensitivities needed in Newton iterations [11, 12, 13, 14, 15]. That line of work is important for constructing matrix rows for constraints and reaction forces, and it also supplies a public-code benchmark suite covering the single pendulum, double pendulum, slider-crank, four-link, and scaling tests. The present paper uses the same body-frame perturbation viewpoint to build a stage system in which the constraint row, velocity row, and acceleration row are all part of the accepted residual; it cannot be read as a fair external comparison until those published benchmark definitions are matched exactly.

Third, high-order collocation and index-reduction methods are a standard route to high-order DAE integration when the constrained flow is regular and the algebraic variables are solved consistently [4, 3]. Symplectic partitioned Runge–Kutta methods also show how holonomic constraints can be preserved within a structure-preserving Runge–Kutta framework [16]. This places the

present construction near constrained collocation, but with a different emphasis on the lower-pair residual rows that are exposed to Newton’s method. The three-stage Gauss path therefore provides a natural sixth-order reference construction on the smooth reduced problem. What is less standard in a lower-pair multibody setting is the concrete residual organization: the accepted path here couples Gauss stage kinematics, Newton–Euler balance, Lie retraction variables, and FullVA lower-pair rows into a square nonlinear system. This is the level at which the manuscript’s contribution is intended to sit; it is not a claim that Gauss collocation itself is new.

Fourth, variational and symplectic integrators derive the discrete equations from a discrete action principle and are valued for their momentum, symplectic, and long-time energy behavior [17, 18, 19]. Constrained variational integrators are therefore a close conceptual neighbor: they treat holonomic constraints at the discrete level and can reduce the constrained solve through null-space coordinates. Their objective, however, is not the same as the FullVA residual used here. The present method does not construct a discrete variational principle; it builds an absolute-coordinate stage residual that enforces lower-pair position, velocity, and acceleration consistency simultaneously and then evaluates the resulting one-step map under the validation policy below.

Fifth, nonsmooth contact and friction solvers form a separate branch of multibody dynamics. Time-stepping and complementarity formulations handle unilateral contact, impact, and Coulomb friction without resolving every event explicitly [20, 21, 22]. Those methods are essential for large contact-rich systems, but their central problem is a contact complementarity or cone-complementarity solve. The validation problems studied here are bilateral lower-pair mechanisms with smooth driving and reaction reconstruction; the paper therefore does not claim a new nonsmooth contact integrator or a replacement for complementarity-based frictional dynamics.

Sixth, time finite elements provide a different route to high-order direct integration. Early space-time and time finite-element methods were developed for structural dynamics and constrained rigid-body dynamics [23, 24, 25]. Later direct integration families based on time finite elements and weighted residuals showed high-order behavior and controllable numerical damping [2, 26, 27]. The frictional Lie-group time finite-element work of Chaturvedi, Sandu, and Sandu extends this direction to index-3 multibody DAEs with nonlinear friction and provides the local paper-style temporal finite-element comparator used here [1]. Its Gauss–Lobatto family uses the order conven-

tion  $2m - 1$ , so its  $m = 3$  member is an order-five formula target, although DAE order reduction can still appear in the reported pendulum experiments.

The present manuscript is intentionally different from a full source-paper reimplementation. It reports method-side evidence for the accepted Gauss collocation FullVA residual on lower-pair mechanisms and compares its conditional sixth-order path against the local order-five temporal finite-element formula target. Complete substitution of all source-paper temporal finite-element weak rows remains outside the accepted claim. The gap addressed here is narrower but concrete: an implemented Lie-group lower-pair stage residual that carries position, velocity, and acceleration consistency through the same nonlinear solve and is checked on the cylindrical-chain benchmark plus four ASME-style mechanisms, with dynamic-order, mechanism-coverage, and diagnostic evidence kept separate.

### 3. Mathematical setting and definitions

Consider a constrained multibody system with generalized position  $q$ , velocity  $v$ , acceleration  $a$ , and Lagrange multipliers  $\lambda$ . The absolute-coordinate index-3 form may be written abstractly as

$$M(q)\dot{v} + G(q, v) + \Phi_q(q)^T \lambda = F(q, v, t), \quad (1)$$

$$\dot{q} = \mathcal{K}(q)v, \quad (2)$$

$$\Phi(q) = 0. \quad (3)$$

Equations (1), (2), and (3) fix the abstract constrained dynamics, kinematics, and holonomic constraint notation used throughout the paper. Here  $\mathcal{K}$  includes the local Lie-group kinematic map for finite rotations. For orientations, the step update is represented by a retraction

$$Q_{n+1} = Q_n \text{Exp}(\eta_{n+1}), \quad (4)$$

with  $\eta_{n+1}$  in the local rotation algebra; Eq. (4) is also the form of the stage rotations  $Q_i = Q_n \text{Exp}(\eta_i)$  used below. The implementation uses this local representation to keep finite rotations on the group while solving a Newton system in vector coordinates.

**Definition 1** (FullVA lower-pair residual). *In this manuscript, FullVA denotes full position-, velocity-, and acceleration-level lower-pair enforcement.*

*The FullVA residual is the square nonlinear stage system that enforces lower-pair position constraints, their velocity consistency equations, and their acceleration consistency equations in the same Newton solve, together with the collocation dynamics and the Lie-group update equations.*

**Definition 2** (Accepted method path). *The accepted method path is the Gauss6/FullVA one-step map generated by the 132-row FullVA residual in the implemented residual path. This is the only method path used for the accepted dynamic-order rows and four-example mechanism-coverage validation claims in this manuscript.*

**Definition 3** (Full-TFE stage replacement). *In this study, full-TFE stage replacement means replacing all 132 Gauss/FullVA stage rows inside Newton by paper-derived temporal finite-element weak rows, without endpoint source data, terminal-row replacement, output projection, or target-direction oracle information. This is a stronger source-paper reproduction criterion and is not accepted by the evidence reported here.*

#### 4. Nomenclature

#### 5. Accepted one-step integrator

Let  $x_n = (q_n, v_n, a_n)$  denote the endpoint position, velocity, and acceleration variables. The accepted method defines a one-step map

$$x_{n+1} = \Psi_h^{\text{G6FVA}}(x_n), \quad (5)$$

Equation (5) is the method-level map induced by one accepted FullVA stage solve, where the stage variables  $Z$  solve the nonlinear residual

$$R_{\text{G6FVA}}(q_n, v_n, a_n, Z; h) = 0. \quad (6)$$

For the cylindrical-chain benchmark used for the main convergence evidence, the residual has two bodies, two lower-pair joints, and three Gauss stages. Each body stage stores translational position, local rotation increment, translational velocity, angular velocity, translational acceleration, and angular acceleration. This gives 18 body unknowns per body per stage. Each lower-pair joint contributes four multiplier unknowns per stage. Thus the Newton stage vector has

$$3\{2(18) + 2(4)\} = 132 \quad (7)$$

Table 2: Main symbols used in the manuscript.

| Symbol               | Meaning                                                                 |
|----------------------|-------------------------------------------------------------------------|
| $q, v, a$            | Generalized position, velocity, and acceleration variables.             |
| $Q$                  | Rotation variable on the Lie group.                                     |
| $\eta$               | Local tangent-space rotation increment used in the retraction.          |
| $\lambda$            | Lagrange multipliers for the lower-pair constraints.                    |
| $M(q)$               | Configuration-dependent mass and inertia operator.                      |
| $F(q, v, t)$         | Applied and generalized force vector.                                   |
| $\Phi(q), \Phi_q(q)$ | Constraint function and constraint Jacobian.                            |
| $\mathcal{K}(q)$     | Kinematic map from generalized velocity to configuration rate.          |
| $h$                  | Time-step size.                                                         |
| $Z$                  | Stacked stage unknown vector in the nonlinear one-step solve.           |
| $R_{\text{G6FVA}}$   | Accepted Gauss6/FullVA residual solved inside Newton.                   |
| $m$                  | Polynomial index for the paper-style Gauss–Lobatto TFE comparator.      |
| $p_{\text{TFE}}$     | Expected order of the local paper-style temporal finite-element target. |

scalar unknowns by Eq. (7), and the accepted residual is square. The closed-loop four-link and slider-crank examples use their own fully constrained kinematic FullVA systems, but they preserve the same position, velocity, and acceleration-level lower-pair residual vocabulary.

The residual is assembled from three row families: lower-pair constraint rows at position, velocity, and acceleration level, Gauss collocation rows for the joint coordinates, and Newton–Euler balance rows. For stage  $i = 1, 2, 3$  with Gauss abscissa  $c_i$ , weights  $b_i$ , and Runge–Kutta matrix  $A_{ij}$ , the stage variables are denoted in Eq. (8):

$$Z_i = (r_i, \eta_i, v_i, \omega_i, a_i, \alpha_i, \lambda_i), \quad (8)$$

where  $r_i$  is the translational position,  $\eta_i$  is the local Lie-algebra rotation increment,  $v_i, \omega_i$  are translational and angular velocities,  $a_i, \alpha_i$  are translational and angular accelerations, and  $\lambda_i$  collects lower-pair multipliers. The implementation evaluates rotations by  $Q_i = Q_n \text{Exp}(\eta_i)$  and uses the right-trivialized differential of the exponential map in the rotational endpoint reconstruction.

*Lower-pair data and joint coordinates.* For lower pair  $j$  let  $P_j(q)$  and  $C_j(q)$  be the parent- and child-side attachment points,  $d_j(q) = C_j(q) - P_j(q)$  the relative displacement,  $a_j(q)$  the parent-fixed joint axis ( $a_0$  is the ground axis and  $a_1 = Q_0 \hat{a}_0^{\text{next}}$ ),  $\tilde{a}_j(q)$  the child-fixed axis, and  $\omega_j^{\text{rel}} = Q_{c(j)} \omega_{c(j)} - Q_{p(j)} \omega_{p(j)}$  the relative angular velocity in the world frame. Each pair carries a fixed world direction  $n_j$ , the joint axis at  $t = 0$ , and a fixed orthonormal pair  $e_{j,1}, e_{j,2} \perp n_j$ . The two joint coordinates of pair  $j$  are the sliding coordinate and the twist angle in Eq. (9),

$$s_j(q) = n_j \cdot d_j(q), \quad \theta_j(q) = \text{atan2}(\tilde{t}_j \cdot (a_j \times t_j), \tilde{t}_j \cdot t_j), \quad (9)$$

where  $t_j, \tilde{t}_j$  are the parent- and child-fixed twist reference vectors. Their rates are  $\dot{s}_j = n_j \cdot \dot{d}_j$  and  $\dot{\theta}_j = a_j \cdot \omega_j^{\text{rel}}$ , and their second rates are  $\ddot{s}_j = n_j \cdot \ddot{d}_j$  and  $\ddot{\theta}_j = a_j \cdot \dot{\omega}_j^{\text{rel}} + \dot{a}_j \cdot \omega_j^{\text{rel}}$ . In the cylindrical-chain benchmark the two joint axes fixed in body 0 are not parallel and body 0 spins, so the axis  $a_1(q)$  of the second pair moves on a cone about the ground axis while  $n_1$  and  $e_{1,m}$  stay frozen at their  $t = 0$  values. The second pair is therefore a four-constraint lower pair with a fixed sliding direction and a moving rotation axis, not a cylindrical pair in the strict sense; its friction sliding speed is  $\dot{s}_1 (n_1 \cdot a_1)$ .

*Stage system.* Write  $\psi_j(q) = \tilde{a}_j(q) - a_j(q)$  for the axis-alignment residual and, for a joint-coordinate function  $g$  with rate  $\dot{g}$ , write

$$\Delta_i[g] = g(Z_i) - g(x_n) - h \sum_{k=1}^3 A_{ik} \dot{g}(Z_k) \quad (10)$$

for its three-stage Gauss collocation defect at stage  $i$ . With  $t_i = t_n + c_i h$ ,  $F_{b,i}^r = f_{b,i} + \Phi_{r,b}^T \lambda_i$ , and  $T_{b,i}^\theta = \tau_{b,i} + \Phi_{\theta,b}^T \lambda_i$ , each stage solves, for every pair  $j$ , every  $m = 1, 2$ , and every body  $b$ , the system in Eq. (11):

$$\begin{aligned} 0 &= e_{j,m} \cdot d_j(q_i), & 0 &= e_{j,m} \cdot \psi_j(q_i), \\ 0 &= e_{j,m} \cdot \dot{d}_j(q_i, v_i), & 0 &= e_{j,m} \cdot \dot{\psi}_j(q_i, v_i), \\ 0 &= e_{j,m} \cdot \ddot{d}_j(q_i, v_i, a_i), & 0 &= e_{j,m} \cdot \ddot{\psi}_j(q_i, v_i, a_i), \\ 0 &= \Delta_i[s_j], & 0 &= \Delta_i[\dot{s}_j], \\ 0 &= \Delta_i[\theta_j], & 0 &= \Delta_i[\dot{\theta}_j], \\ 0 &= m_b a_{b,i} - F_{b,i}^r, & 0 &= J_b \alpha_{b,i} + \omega_{b,i} \times J_b \omega_{b,i} - T_{b,i}^\theta. \end{aligned} \quad (11)$$

The first three lines are the lower-pair constraints at position, velocity, and acceleration level (12 rows per pair and stage); the fourth and fifth lines are three-stage Gauss collocation of the four joint coordinates  $(s_j, \dot{s}_j, \theta_j, \dot{\theta}_j)$  through the defect operator of Eq. (10) (4 rows per pair and stage); the last line is the Newton–Euler balance (6 rows per body and stage). Per stage this gives  $2 \times (12 + 4) + 2 \times 6 = 44$  rows for the 44 unknowns of Eq. (7): 72 constraint rows, 24 joint-coordinate collocation rows, and 36 Newton–Euler rows over the three stages. The absolute body coordinates  $(r_i, \eta_i, v_i, \omega_i)$  carry no collocation row of their own; they are determined by the constraint rows once the joint coordinates are collocated. The implementation projects the two translational collocation rows onto the current joint axis  $a_j(q_i)$  rather than onto  $n_j$ ; on the constraint manifold  $d_j, \dot{d}_j, \ddot{d}_j \parallel n_j$ , so the two forms differ by the nonzero factor  $n_j \cdot a_j(q_i)$  and define the same stage solution. Driven mechanisms add prescribed right-hand sides  $g_i^{(0)}, g_i^{(1)}, g_i^{(2)}$  to the three constraint levels. The Newton–Euler rows are, for each body  $b$ ,

$$m_b a_{b,i} - f_{b,i} - \Phi_{r,b}(q_i)^T \lambda_i = 0, \quad (12)$$

$$J_b \alpha_{b,i} + \omega_{b,i} \times J_b \omega_{b,i} - \tau_{b,i} - \Phi_{\theta,b}(q_i)^T \lambda_i = 0, \quad (13)$$

where the multiplier wrench collects the joint normal forces, the axis alignment torques, and the regularized Brown–McPhee friction force along the

joint axis. The lower-pair constraint block, written compactly as

$$\Phi(q) = 0, \quad \Phi_q(q)v = 0, \quad \Phi_q(q)a + \dot{\Phi}_q(q, v)v = 0, \quad (14)$$

is enforced at all three levels at every stage; Eq. (14) is the compact form of the first three lines of Eq. (11). The lower-pair library used for the four ASME-style examples covers coordinate-difference (CD), dot-product-one (DP1), dot-product-two (DP2), and distance (D) constraints; the closed-loop examples use their own fully constrained kinematic FullVA systems with the same three-level vocabulary. The method is therefore a FullVA constrained stage solve, not a post-step position-only projection.

After Newton convergence, the endpoint is reconstructed by the Gauss weights,

$$r_{n+1} = r_n + h \sum_{i=1}^3 b_i v_i, \quad v_{n+1} = v_n + h \sum_{i=1}^3 b_i a_i, \quad (15)$$

$$Q_{n+1} = Q_n \text{Exp} \left( h \sum_{i=1}^3 b_i J_r^{-1}(\eta_i) \omega_i \right), \quad \omega_{n+1} = \omega_n + h \sum_{i=1}^3 b_i \alpha_i. \quad (16)$$

The Newton linearization solves

$$\frac{\partial R_{\text{G6FVA}}}{\partial Z} \Delta Z = -R_{\text{G6FVA}}, \quad (17)$$

with automatic differentiation used to assemble the dense reference Jacobian. The sparse backend uses the same residual and Jacobian entries; it is tracked as an implementation caveat rather than as a primary accuracy claim.

### 5.1. Complete execution algorithm

Table 3 summarizes one accepted step; the implemented Gauss6/FullVA path used for the cylindrical-chain and four-example validation rows is the trajectory algorithm below.

The listing is deliberately written as an execution protocol: it starts from mechanism data and ends with the error/order records used in the reproduced results.

Operationally, for an example  $e$  and a step size  $h$ , the accepted run is the

Table 3: Algorithmic form of one Gauss6/FullVA step.

| Step | Operation                                                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Build the lower-pair constraint graph, mass/inertia blocks, loads, and FullVA stage unknown vector for $(q_n, v_n, a_n)$ and step size $h$ .                                                                                   |
| 2    | Assemble the 132-row residual $R_{\text{G6FVA}}(q_n, v_n, a_n, Z; h)$ of Eq. (11): lower-pair constraint rows at position, velocity, and acceleration level, joint-coordinate collocation rows, and Newton–Euler balance rows. |
| 3    | Solve the nonlinear stage system by Newton iteration with automatic differentiation for the residual Jacobian.                                                                                                                 |
| 4    | Recover the endpoint by Gauss quadrature and the Lie-group retraction in Eq. (4), then apply the endpoint closure map.                                                                                                         |
| 5    | Check endpoint position, velocity, residual, rank, order, and four-example quantities using the reproducibility checks summarized in the reproducibility appendix.                                                             |

composition in Eq. (18):

$$\begin{aligned}
Z_{n,h,e} &= \text{NewtonSolve}_{\tau_N, k_{\max}} \left( R_{\text{G6FVA}}(x_{n,h,e}, Z; h) = 0; Z_{n,h,e}^{(0)} \right), \\
\tilde{x}_{n+1,h,e} &= \mathcal{E}_h(x_{n,h,e}, Z_{n,h,e}), \\
x_{n+1,h,e} &= \mathcal{C}_h(\tilde{x}_{n+1,h,e}), \\
\mathcal{R}_e(h) &= \mathcal{P}_e \left( \{x_{n,h,e}\}_{n=0}^{N_h}, \{D_{n,h,e}\}_{n=0}^{N_h-1}, x_e^{\text{ref}} \right).
\end{aligned} \tag{18}$$

Here  $\mathcal{E}_h$  is the endpoint reconstruction in Eqs. (15)–(16),  $\mathcal{C}_h$  is the accepted endpoint closure map,  $D_{n,h,e}$  stores Newton and constraint diagnostics,  $x_e^{\text{ref}}$  is the analytic or nested local reference specified by the validation policy, and  $\mathcal{P}_e$  is the postprocessing map that writes the reproduced error, order, diagnostic, and comparison-scope records.

**Algorithm 1 (complete accepted Gauss6/FullVA run).**

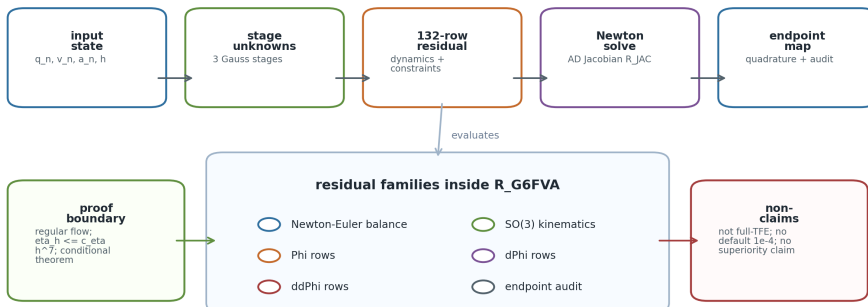
1. Choose the validation example  $e$ , its mechanism data, its step-size set  $H_e$ , final time  $T_e$ , Newton tolerance  $\tau_N$ , Newton cap  $k_{\max}$ , Jacobian backend, and reference policy.
2. Build the body mass and inertia blocks, lower-pair graph, joint frames, axes, CD/DP1/DP2/D constraint bases, driven right-hand sides, loads, and friction data.
3. Construct the initial state  $x_{0,h,e}$  on  $\mathbb{R}^3 \times \text{SO}(3)$  for every body and project the initial velocity onto the velocity-level lower-pair constraints.
4. For each  $h \in H_e$ , first check the accepted-grid condition  $|T_e/h - \text{round}(T_e/h)| \leq \varepsilon_{\text{grid}}$ . Accepted order rows use only grids satisfying this condition; non-integer final-time grids are recorded as external-comparison diagnostics rather than promoted to reported order rows. This is a validation/reporting policy for the finite order rows, not an additional theorem condition; the proof theorem below states its grid-point estimate on the reported grid and does not require exact final-time divisibility. Set  $N_h = \text{round}(T_e/h)$ , precompute the three-stage Gauss coefficients and the fixed FullVA stage-vector layout, and initialize empty trajectory and diagnostic records.
5. For  $n = 0, \dots, N_h - 1$ , form the predictor  $Z_{n,h,e}^{(0)}$  from the endpoint state, constant-acceleration motion, zero initial Lie increments, zero angular-velocity and angular-acceleration guesses, and force-balance multiplier estimates.

6. Assemble  $R_{\text{G6FVA}}(x_{n,h,e}, Z; h)$  by evaluating stage poses, lower-pair positions, velocities, and accelerations, the FullVA position-, velocity-, and acceleration-level constraint rows, the joint-coordinate collocation rows, and the Newton–Euler balance rows.
7. Iterate Newton: evaluate the residual, form or apply  $\partial R_{\text{G6FVA}}/\partial Z$ , solve Eq. (17), update  $Z$ , and stop only after the residual or update norm satisfies  $\tau_N$ ; reject the step if  $k_{\max}$  is reached.
8. Reconstruct  $\tilde{x}_{n+1,h,e}$  with the Gauss endpoint formulas and the Lie-group retraction, then evaluate raw endpoint position and velocity constraint residuals.
9. Apply the accepted endpoint closure map  $\mathcal{C}_h$  when the velocity-level closure threshold is exceeded, and record raw residuals, corrected residuals, projection rank, condition number, and correction norm.
10. Store  $x_{n+1,h,e}$  and the diagnostics  $D_{n,h,e}$ , including Newton iterations, residual norms, linear-solve residuals, Jacobian or colored-derivative timings, endpoint constraint norms, stage acceleration constraint norms, and quaternion unit error.
11. After all steps and all  $h \in H_e$  are complete, compare each trajectory with  $x_e^{\text{ref}}$ , compute the paper’s error norms, error ratios, observed orders, mechanism-coverage rows, and reaction residual summaries.
12. Write the CSV, JSON, plot, metadata, and comparison-scope records used in the manuscript. These postprocessing checks reproduce the evidence boundary; they do not alter the accepted numerical trajectory.

All rows in Algorithm 1 are the accepted Gauss/FullVA rows in Eq. (6). A full source-paper TFE replacement would replace the stage residual assembled in the residual-assembly step by paper-derived weak rows; that stronger algorithm is explicitly outside the accepted claim in this manuscript. Algorithm–theorem tolerance boundary: Algorithm 1 records the executable finite-run protocol, including the reported Newton tolerance  $\tau_N$ . Theorem 1 uses the same implemented residual map only after the accepted branch also satisfies the scaled proof policy  $\eta_h^{\text{tube}} \leq c_\eta h^7$  in the fixed proof norm. A fixed  $\tau_N$  run is therefore theorem-admissible only for those steps on which its accepted branch-selected residuals meet that  $h^7$  bound; otherwise it is finite-run diagnostic evidence and does not discharge P6. The correspondence between Algorithm 1 and the supplied source package is summarized in Appendix Table B.31; that table is a reproducibility map, not an additional numerical claim.

### Accepted Gauss6/FullVA one-step architecture

The figure separates the mathematical one-step map from open source-paper TFE replacement and external-superiority claims.



Accepted claim: a conditional sixth-order Gauss6/FullVA path for lower-pair mechanisms. Open gates remain explicit rather than hidden in the method diagram.

Figure 1: Accepted Gauss6/FullVA one-step architecture. The schematic separates the 132-row FullVA residual and conditional order boundary from non-claims: it is not a full-TFE replacement, does not require a default  $10^{-4}$  step-size sweep, and does not assert external superiority without same-test evidence.

The reproducibility record is described in the reproducibility appendix. The main text uses that record only to keep the claim boundary precise; the scientific argument rests on the residual definition, the conditional sixth-order theorem, and the numerical evidence below.

#### Formal-order comparator guardrail.

This short guardrail fixes only the scalar formal-order comparator used later in the paper. The comparison to Lie-group TFE methods is formal-order-level unless same-test implementation evidence is provided. The source-paper comparison used in this manuscript is therefore local and explicit. It is placed before validation only to define the comparator convention; it is not benchmark evidence, not a theorem input, and not a claim that the TFE implementation in the source paper has been reproduced. The main evidence chain remains method definition, conditional FullVA proof, accepted dynamic-order rows, and mechanism-coverage rows. The paper-style temporal finite-element family indexed by polynomial degree  $m$  uses the expected order convention

$$p_{\text{TFE}}(m) = 2m - 1. \quad (19)$$

The  $m = 3$  Gauss–Lobatto temporal finite-element target therefore has expected order five. The source paper should not be read as reporting uniform fifth-order DAE convergence in every experiment: in its pendulum study it observes that the  $m = 3$  TFE implementation can lose order on the DAE problem, even though the underlying Kim–Reddy TFE construction has the expected  $2m - 1$  formula order [2, 1]. This manuscript therefore uses the paper-style formula target  $p_{\text{TFE}}(3) = 5$  as the comparator boundary, not a weakened observed-order target. The accepted Gauss6/FullVA path has a conditional method-order claim of six. The manuscript-level order comparison is the formal-order comparison of that conditional Gauss6/FullVA path against the local order-five target. It is not a claim that the present implementation is faster or more accurate than the source-paper TFE implementation on the original source-paper tests.

**External-comparison guardrail.**

The comparison policy used in this paper is deliberately conservative. A statement about implemented accuracy or cost against a source-paper method is accepted only after a same-test row satisfies all entries in Table 5. Until then, the TFE comparison remains the formal-order comparison in Eq. (19), and work/precision plots are candidate or common-reference diagnostics.

Consequently the manuscript does not say that Gauss6/FullVA is faster or more accurate than an implemented source-paper TFE method. The stronger sentence would require the Table 5 promotion criterion to close on the same benchmark. The accepted claim is narrower: Gauss6/FullVA is a conditional sixth-order Lie-group Gauss-collocation FullVA path under the retained branch-smoothness, endpoint/stability, and solver-scale interfaces later stated in Theorem 1.

## 6. Validation protocol

The validation protocol organizes the numerical evidence into three layers: accepted dynamic-order evidence for the Gauss6/FullVA branch, mechanism-coverage evidence for lower-pair examples, and diagnostic comparison/reproducibility boundaries. It also fixes the error norms and reference policies used in the paper, so that the numerical tables are not merely unsupported assertions. For a trajectory sampled at accepted output times  $t_k$ , translational errors use the maximum-in-time Euclidean or infinity norms recorded in the CSV files. Orientation errors use the Lie-group

Table 4: Comparison boundary used in the manuscript.

| Question                             | Boundary                                                                                                                                                                                                                                                                                                                           |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| What is claimed?                     | Under the stated compact-branch, endpoint/stability, and solver-scale contracts, Gauss6/FullVA has a conditional sixth-order Gauss-collocation FullVA path. Its formal-order contrast is order six versus the local order-five $m = 3$ Gauss–Lobatto TFE formula target; no implementation superiority follows from this contrast. |
| What is the comparator?              | A local paper-style temporal finite-element formula target with expected order $2m - 1 = 5$ ; source-paper implementation superiority requires same-test evidence.                                                                                                                                                                 |
| What is not claimed?                 | Complete source-paper residual reproduction and accepted independent full-TFE stage replacement; faster or more accurate implemented source-paper TFE performance on the original tests is also not claimed.                                                                                                                       |
| Where is the reproducibility record? | The reproducibility appendix lists the source package, comparison-scope files, and read-only checks used to reproduce the reported evidence.                                                                                                                                                                                       |

Table 5: Guardrail for promoting a baseline row to an implemented same-test comparison.

| Promotion item           | Required evidence                                                                                                                                                   | Evidence boundary                          |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| Mechanism and parameters | The source-paper frictionless pendulum or revolute-friction benchmark is run with the same masses, geometry, loads, friction law, final time, and output variables. | Open for external same-test promotion.     |
| Methods                  | The same script or verified harness reports Newmark- $\beta$ , trapezoidal, TFE $m = 1, 2, 3$ , and Gauss6/FullVA rows.                                             | Candidate diagnostics only.                |
| Grid and solver policy   | The rows use the same step sizes, same reference solution, same Newton tolerance policy, and same accepted-grid handling.                                           | Not fully same-test closed.                |
| Error norms              | Position, velocity, orientation, angular-velocity, constraint, and reaction or multiplier errors are evaluated with the same norm definitions.                      | Partly available in common-reference rows. |
| Work metric              | Wall time, Newton iterations, Jacobian assembly time, and linear-solve time are reported so that an accuracy claim is not separated from cost.                      | Diagnostic compendium only.                |

distance

$$d_R(Q_h, Q_{\text{ref}}) = \|\log(Q_{\text{ref}}^T Q_h)\|_2. \quad (20)$$

Equation (20) defines the rotation component of the reported trajectory error. Angular-velocity errors use the corresponding vector norm. The observed order reported in the tables is the least-squares slope of  $\log e(h)$  against  $\log h$  over the stated step sizes. The fit protocol is fixed before the reported row is inspected: step window, reference policy, reported components, error norm, and inclusion or exclusion of floor-limited rows are the policy stated for that evidence row. Recomputing a slope after dropping a coarse or fine row, swapping position, velocity, or orientation components, changing a norm, or moving a row between dynamic-order, mechanism-coverage, and diagnostic status defines a different evidence row. Such a post-hoc row may be recorded as a new diagnostic, but it cannot upgrade the theorem claim or replace the predeclared row in the acceptance policy. A result can support the main manuscript claim only if it satisfies the following acceptance criteria.

1. The method identifier is Gauss6/FullVA, and the order claim is the conditional sixth-order path of Theorem 1, not an unconditional property of fixed-tolerance or source-policy rows.
2. The smooth cylindrical-chain h-sweep uses at least three step sizes and reports position and velocity orders above the local order-five comparator.
3. The four ASME-style examples are accepted under the same method-side policy, but their roles differ: only the single and double pendula are used as accepted dynamic-order examples, while the closed-loop examples provide mechanism coverage, constraint closure, reaction residuals, and local coarse dynamic evidence.
4. The source-paper comparator remains a local  $m = 3$  Gauss-Lobatto temporal finite-element target with expected order five.
5. Any claim against a source-paper TFE implementation requires same-test evidence with matched parameters, step sizes, tolerances, reference solution, error norm, and work metric; this manuscript does not make that source-paper superiority claim.
6. Complete source-paper residual reproduction and independent full-TFE stage replacement remain outside the accepted method claim.
7. Routine reproducibility checks are documented separately in the reproducibility appendix.

Table 6: Reproducibility policy for the numerical evidence.

| Evidence block                   | Step sizes                               | Reference                                          | Primary reported quantities                                        |
|----------------------------------|------------------------------------------|----------------------------------------------------|--------------------------------------------------------------------|
| Smooth cylindrical chain         | 0.04, 0.02, 0.01                         | $h = 0.005$ trajectory                             | Pos./vel. errors; constraints; Newton iterations.                  |
| Sharp-friction cylindrical chain | 0.04, 0.02, 0.01 and refined diagnostics | $h = 0.005$ or finer reference by diagnostic check | Order reduction, endpoint constraints, refinement cost.            |
| Driven single pendulum           | 0.2, 0.1, 0.05                           | Analytic $\theta(t)$ trajectory                    | Position, velocity, orientation, and residual errors.              |
| Double pendulum                  | 0.04, 0.02, 0.01                         | Nested local FullVA $h = 0.005$ reference          | Orientation and angular-velocity errors plus endpoint constraints. |
| Four-link and slider-crank       | 0.02, 0.01, 0.005                        | Local kinematic FullVA $h = 0.001$ reference       | Constraints; rank; reaction residuals.                             |

The paper uses three evidence types throughout the results. A dynamic-order row is used to support an order statement for the accepted one-step map. A mechanism-coverage row checks that the same FullVA position, velocity, and acceleration vocabulary closes a lower-pair mechanism and produces consistent reaction dynamics, but it is not by itself an asymptotic order row. A diagnostic row clarifies a boundary condition, failure mode, or external-comparison gap and is never promoted to a theorem or superiority claim. This separation is important for the four-link and slider-crank examples, where the closed-loop constraint/reaction checks and coarse-window trajectory traces are reported only as coverage/consistency diagnostics; they do not become accepted asymptotic order proof, residual-to-error evidence, or source-policy comparison closure.

The accepted cylindrical-chain h-sweep is reported with endpoint velocity closure, and the closure itself is checked explicitly rather than hidden inside the error table. Two endpoint records are retained. The first is the actual projected endpoint path used for the main finite-window order row. The second is a square endpoint KKT check in which endpoint velocity variables, least-squares stationarity, and endpoint velocity constraints are solved as an augmented closure residual. Table 7 reports the smooth branch scaling. The endpoint functional  $E$  used in Lemma 5 is the velocity-level/KKT closure subsystem actually corrected by  $\mathcal{C}_h$ , not every raw endpoint diagnostic in the table. The raw endpoint position defect is retained as a separate constraint monitor: it remains small and approximately sixth order over the reported window, but it is not the  $\|E(z_u)\| \leq C_{E,\text{raw}} h^7$  hypothesis. The raw endpoint velocity defect and the KKT correction norm both have finite-window fitted slopes near seven over the reported window, and the corrected endpoint velocity residual is at roundoff scale. This evidence is a finite-run consistency check for the endpoint-closure perturbation scale in Lemma 5; it is not a proof of that lemma’s retained P2 endpoint-ball, raw-defect, or uniform right-inverse hypotheses, and it neither closes the full-TFE replacement nor the source-policy boundaries. The sharp-branch coarse h-sweep remains a limitation, not theorem evidence.

## 7. Conditional sixth-order theorem

This section proves the conditional sixth-order estimate for the accepted Gauss6/FullVA map. The argument has three steps. First, Lemma 1 shows that the implemented stage system of Eq. (11) is exactly three-stage Gauss

Table 7: Smooth cylindrical-chain endpoint closure scaling diagnostic.

| $h$        | Raw pos. defect         | Raw vel. defect         | KKT correction          | Closed vel. defect      | Newton iters. |
|------------|-------------------------|-------------------------|-------------------------|-------------------------|---------------|
| 0.04       | $5.883 \times 10^{-12}$ | $9.517 \times 10^{-11}$ | $7.926 \times 10^{-11}$ | $2.431 \times 10^{-16}$ | 12            |
| 0.02       | $9.196 \times 10^{-14}$ | $7.427 \times 10^{-13}$ | $6.395 \times 10^{-13}$ | $1.573 \times 10^{-16}$ | 24            |
| 0.01       | $1.635 \times 10^{-15}$ | $6.505 \times 10^{-15}$ | $5.484 \times 10^{-15}$ | $3.634 \times 10^{-16}$ | 48            |
| Obs. order | 5.907                   | 6.918                   | 6.909                   | roundoff limited        | –             |

collocation of the reduced joint-coordinate equation of motion, written in absolute coordinates; the stage residual therefore vanishes identically at the lifted reduced Gauss stage and no stage-residual perturbation term appears. Second, the two implementation-side perturbations that do appear, endpoint closure and inexact Newton, are bounded by  $O(h^7)$  under explicit local hypotheses (Lemmas 5 and 6). Third, the local defect is transferred to the reported grid by a discrete Gronwall argument with tube retention (Lemma 8). The retained hypotheses are collected in Assumption 1; they fix the compact branch, the local inverses, the stability scale, and the solver envelope before any numerical evidence is read. The algebraic identities and the perturbation lemmas below are machine-checked in a Lean 4/Mathlib development distributed with the source package (Appendix B); Butcher’s order theorem for the Gauss tableau and the classical Gauss local-error estimate [3] are used as cited results.

### 7.1. Reduced chart, lift, and retained hypotheses

Let  $y = (s_0, \theta_0, s_1, \theta_1, \dot{s}_0, \dot{\theta}_0, \dot{s}_1, \dot{\theta}_1)$  collect the joint coordinates of Eq. (9) and their rates. On the smooth constrained branch the equations of motion reduce to

$$\dot{y} = f(y, t), \quad (21)$$

where  $f$  returns the rates and the joint accelerations obtained from the constrained Newton–Euler equations. Let  $\varphi_h$  denote the flow of Eq. (21) over one step,  $\mathcal{L}(y, t) = (q, v, a, \lambda)$  the smooth lift to absolute coordinates given by forward kinematics with the velocities, accelerations, and multipliers of the constrained dynamics, and  $Y_1, Y_2, Y_3$  the three-stage Gauss collocation stages of Eq. (21) from  $y_n$ , so that

$$Y_i = y_n + h \sum_{k=1}^3 A_{ik} f(Y_k, t_k), \quad i = 1, 2, 3. \quad (22)$$

The lifted reduced Gauss stage is  $Z_G = (\mathcal{L}(Y_1), \mathcal{L}(Y_2), \mathcal{L}(Y_3))$ . We write  $F_{A,h}(\cdot; x_n)$  for the implemented row-scaled 132-row residual of Eq. (11),  $\mathcal{E}_h$  for the Gauss-weight endpoint reconstruction of Eqs. (15)–(16),  $\mathcal{C}_h$  for the accepted endpoint-closure map,  $\mathcal{P}_h = \mathcal{C}_h \circ \mathcal{E}_h$ , and  $\Psi_h^G = \mathcal{E}_h(Z_G)$  for the reconstructed Gauss endpoint. All norms below are fixed finite-dimensional norms on the reduced chart, the stage space, and the endpoint space; chart-to-configuration norm equivalences on the compact tube are absorbed into the constants.

**Assumption 1** (Accepted-path regularity and perturbation scale). *Fix a final time  $T$  and a compact reduced-state proof tube  $K$  containing the exact trajectory of Eq. (21) at distance at least  $d_K > 0$  from  $\partial K$ . The following hold uniformly for  $y \in K$ ,  $0 \leq t \leq T$ , and  $0 < h \leq h_0$ , with constants independent of  $h$  and of the reported grid length.*

- P1. (Smooth branch.) The lower-pair constraint Jacobian has constant full row rank, the constrained mass matrix is nonsingular on the tangent space,  $f$  is  $C^7$ , and the lift  $\mathcal{L}$  is smooth on the tube.*
- P2. (Local inverses and stability.) The stage Jacobian  $J_h = D_Z F_{A,h}(Z_G; x_n)$  is invertible with  $\|J_h^{-1}\| \leq M$  and  $\|D_Z^2 F_{A,h}\| \leq L$  on a ball  $B_\rho(Z_G)$  with  $ML\rho \leq \frac{1}{2}$ . The endpoint-closure subsystem  $E$  of Lemma 5 is  $C^2$  on a ball  $B_{r_E}(z_*)$  around a same-branch closed anchor  $z_*$  with  $E(z_*) = 0$ ;  $DE(z_*)$  has a right inverse  $B_*$  with  $\|B_*\| \leq M_{E,\text{ri}}$ ,  $\|DE(z) - DE(z_*)\| \leq L_E\|z - z_*\|$  on that ball,  $M_{E,\text{ri}}L_E r_E \leq \frac{1}{2}$ , the unclosed endpoint  $z_u = \Psi_h^G$  lies in  $B_{r_E/2}(z_*)$ , and its raw closure defect satisfies  $\|E(z_u)\| \leq C_{E,\text{raw}}h^7$ . The accepted one-step map satisfies the stability scale*

$$\|\Psi_h^{\text{G6FVA}}(y) - \Psi_h^{\text{G6FVA}}(\bar{y})\| \leq (1 + C_s h)\|y - \bar{y}\|, \quad y, \bar{y} \in K, \quad (23)$$

with  $C_s \geq 0$ .

- P6. (Solver envelope.) The computed Newton iterate  $\tilde{Z}_A$  is initialized from the Gauss predictor, lies in the ball  $B_\rho(Z_G)$  of P2, and its proof-norm residual satisfies  $\|F_{A,h}(\tilde{Z}_A; x_n)\| \leq \eta_h^{\text{tube}} \leq c_\eta h^7$ . The endpoint-output map satisfies  $\|D_Z \mathcal{P}_h\| \leq M_N$  on that ball.*

P1 and P2 are compact-tube regularity hypotheses in the sense of classical collocation theory; they are not inferred from finite rank probes, residual tables, or observed convergence slopes. P6 is a theorem-level solver policy: a fixed absolute Newton tolerance is a finite-run engineering choice and

supports the asymptotic statement only on transitions where the accepted residual meets the  $h^7$  envelope in the fixed proof norm. The numerical section reports fixed-tolerance runs as finite-run diagnostics of this policy, not as its proof. The theorem also uses the classical Gauss local-error estimate: for  $f \in C^7$  on the compact tube there is  $C_{G,0}$  with

$$\|u(t_n + h) - \varphi_h(y_n)\| \leq C_{G,0}h^7, \quad (24)$$

where  $u$  is the degree-three collocation polynomial through Eq. (22) [3]. The tableau used in the implementation satisfies Butcher’s simplifying assumptions  $B(6)$ ,  $C(3)$ ,  $D(3)$  and fails  $B(7)$ , so its order is exactly six; these algebraic conditions are machine-checked for the coefficients hard-coded in the source package.

### 7.2. Exact stage identity

**Lemma 1** (Exact stage identity at the lifted reduced Gauss stage). *Under P1, the implemented 132-row residual satisfies*

$$F_{A,h}(Z_G; x_n) = 0. \quad (25)$$

*Conversely, on the branch where  $n_j \cdot a_j(q_i) \neq 0$ , every stage vector that satisfies the 96 non-dynamic rows of Eq. (11) lies on the constraint manifold with consistent velocities and accelerations and has joint coordinates satisfying Eq. (22). Hence the accepted stage solve is three-stage Gauss collocation of Eq. (21) in the joint-coordinate chart, written in absolute coordinates.*

*Proof.* The 72 constraint rows vanish on  $Z_G$  because each  $\mathcal{L}(Y_k)$  lies on the constraint manifold with consistent velocities and accelerations. The 24 joint-coordinate rows are, by Eq. (9), the reduced collocation equations (22) read component-wise, because  $s_j, \theta_j, \dot{s}_j, \dot{\theta}_j$  are coordinates of  $y$  and  $\ddot{s}_j, \ddot{\theta}_j$  are components of  $f$ ; the implemented projection onto  $a_j(q_i)$  multiplies each translational row by  $n_j \cdot a_j(q_i)$ . The 36 Newton–Euler rows vanish because  $\mathcal{L}$  supplies the stage accelerations and multipliers satisfying force and moment balance, and the implemented rows (12)–(13) are exactly these balance defects with the implemented wrench sign conventions. For the converse, the constraint rows force  $d_j, \dot{d}_j, \ddot{d}_j \parallel n_j$ , so each translational collocation row factors as  $(s_j(Z_i) - s_j(x_n) - h \sum_k A_{ik} \dot{s}_j(Z_k)) (n_j \cdot a_j(q_i))$ , and the twist rows are already scalar. Both directions are verified row by row for the implemented residual in the accompanying Lean development.  $\square$

Lemma 1 replaces any stage-residual perturbation estimate: the residual value at the lifted Gauss stage is not small, it is zero. What remains to be controlled is which root Newton selects, how the endpoint is reconstructed and closed, and how inexact the Newton solve is. Lemma 1 is stated for the open-chain benchmark, where the joint coordinates form a global chart; for the closed-loop examples the same argument applies on any local chart of independent coordinates, but no dynamic-order claim is made for them.

*Implementation fidelity.* The rows of Eq. (11) are transcribed from the single implemented residual function, `residual_cylindrical_chain`, and the Newton Jacobian `R_JAC` is its automatic derivative, so there is one residual/Jacobian implementation path and no hand-coded Jacobian. A static implementation certificate in the source package keeps the source-identity record of that path, and a runtime formula-row oracle re-evaluates all 132 rows from the formula-level expressions on deterministic probes and compares them with the runtime residual and Jacobian. The Lean development transcribes the same rows and proves Lemma 1 for the transcription; the binding between transcription and source is the static certificate, not a theorem.

**Lemma 2** (Branch selection). *Under P2,  $Z_G$  is the unique zero of  $F_{A,h}(\cdot; x_n)$  in  $B_\rho(Z_G)$ . In particular the exact accepted stage root  $Z_A$  on the Gauss-predictor branch equals  $Z_G$ , and  $\mathcal{E}_h(Z_A) = \Psi_h^G$ .*

*Proof.* For  $Z \in B_\rho(Z_G)$  the mean value inequality gives  $\|F_{A,h}(Z) - F_{A,h}(Z_G) - J_h(Z - Z_G)\| \leq \frac{L\rho}{2} \|Z - Z_G\|$ . If  $F_{A,h}(Z) = 0$ , then Eq. (25) gives the chain Eq. (26),

$$\begin{aligned} \|Z - Z_G\| &= \|J_h^{-1} J_h(Z - Z_G)\| \leq M \|F_{A,h}(Z) - F_{A,h}(Z_G) - J_h(Z - Z_G)\| \\ &\leq \frac{ML\rho}{2} \|Z - Z_G\| \leq \frac{1}{4} \|Z - Z_G\|, \end{aligned} \tag{26}$$

so  $Z = Z_G$ . □

**Lemma 3** (P2 from P1 for small steps). *Assume P1, and assume that on the tube the map  $q \mapsto (\Phi(q), s_0(q), \theta_0(q), s_1(q), \theta_1(q))$  has an invertible Jacobian, i.e. the joint coordinates complete the lower-pair constraints to a chart. Then there is  $h_0 > 0$  such that for  $0 < h \leq h_0$ : (i) the stage Jacobian  $J_h = D_Z F_{A,h}(Z_G)$  is invertible with*

$$\|J_h^{-1}\| \leq 2\|J_0^{-1}\|, \quad J_0 = \lim_{h \rightarrow 0} J_h, \quad \|J_h - J_0\| \leq C_J h, \tag{27}$$

(ii) the endpoint-closure Jacobian is invertible, so  $B_*$  exists with  $\|B_*\|$  bounded uniformly, (iii) the raw endpoint defect satisfies  $\|E(z_u)\| \leq C_{E,\text{raw}}h^7$ , and (iv) the accepted map satisfies the stability scale Eq. (23). Hence, with the smoothness bounds on the compact tube, all of P2 follows from P1 for small steps.

*Proof.* At  $h = 0$  the collocation rows of Eq. (11) reduce to  $s_j(Z_i) - s_j(x_n)$ ,  $\theta_j(Z_i) - \theta_j(x_n)$  and their rate analogues, so  $J_0$  is block diagonal over the stages and block triangular within a stage: the position block is the Jacobian of  $q \mapsto (\Phi, s, \theta)$ , the velocity block is the same matrix acting on  $(v, \omega)$  because on the constraint manifold the rate rows are the time derivatives of the coordinate rows, and the acceleration block is the constrained Newton–Euler system  $\begin{bmatrix} M & \Phi_q^T \\ \Phi_q & 0 \end{bmatrix}$  acting on  $(a, \alpha, \lambda)$ . Each block is invertible under P1 and the chart assumption, so  $J_0$  is invertible with an  $h$ -independent inverse bound. The only  $h$ -dependence of  $F_{A,h}$  is the factor  $h \sum_k A_{ik}$  in the collocation rows, so  $\|J_h - J_0\| \leq C_J h$  on the compact tube. For  $2\|J_0^{-1}\|C_J h \leq 1$  the perturbation bound Eq. (27) follows from the Neumann series; this step is machine-checked (`uniform_inverse_of_perturbation`). The endpoint closure is a square KKT system in the endpoint velocity variables whose matrix is  $\begin{bmatrix} I & \Phi_q^T \\ \Phi_q & 0 \end{bmatrix}$ , invertible by full row rank of  $\Phi_q$ ; its raw defect is the velocity-level constraint value at the Gauss-weight endpoint, which is the quadrature error of Lemma 4 and hence  $O(h^7)$ ; the same implicit-function argument gives a same-branch closed anchor  $z_*$  within  $O(h^7)$  of  $z_u$ , so  $z_u \in B_{r_E/2}(z_*)$  for small  $h$ . Finally, by Lemma 1 the accepted map read in the reduced chart is the reduced Gauss step, whose derivative is  $I + O(h)$  by the implicit function theorem for the stage equations, composed with the smooth endpoint reconstruction and an  $O(h^7)$  closure correction; this gives the Lipschitz scale  $1 + C_s h$ .  $\square$

With Lemma 3, P2 is not an independent hypothesis for the cylindrical-chain benchmark but a consequence of P1 and a small-step threshold; it is kept as a separate item because the theorem is stated for any mechanism whose stage system has the form of Eq. (11). The threshold  $h_0$  of the Neumann argument is not sharp: in the Euclidean norm on the implemented stage layout, Table 12 gives  $h_0 \approx 3 \times 10^{-5}$  on the benchmark, while the conclusion  $\|J_h^{-1}\| \leq 2\|J_0^{-1}\|$  is observed with ratio at most 1.55 on every reported grid.

### 7.3. Endpoint defect, endpoint closure, and inexact Newton

**Lemma 4** (Gauss endpoint defect). *Under P1 there is a constant  $C_G$ , uniform on the compact tube, such that*

$$\|\Psi_h^G(y) - \varphi_h(y)\| \leq C_G h^7. \quad (28)$$

*Proof.* Let  $u$  be the collocation polynomial of Eq. (22) and  $g$  any component of the lifted state along  $u$ , so that  $g(t) = \rho(u(t))$  for a smooth chart function  $\rho$  and  $g'(t_k)$  equals the lifted stage velocity at  $Y_k$ ; for the rotational component,  $g$  is the Lie-algebra increment  $\eta(t)$  with  $\eta' = J_r^{-1}(\eta)\omega$  along the lifted trajectory. The reconstruction Eqs. (15)–(16) replaces  $g(t_n + h) - g(t_n) = \int_{t_n}^{t_n+h} g'(t) dt$  by the Gauss quadrature  $h \sum_i b_i g'(t_i)$ . Because the Gauss weights integrate polynomials of degree at most five exactly, a Peano-kernel argument gives

$$\left| \int_{t_n}^{t_n+h} g'(t) dt - h \sum_{i=1}^3 b_i g'(t_i) \right| \leq \frac{h^7}{60} \max_{[t_n, t_n+h]} |g^{(7)}|, \quad (29)$$

with  $g^{(7)}$  bounded uniformly on the tube by P1 and the smoothness of  $u$ . Hence each reconstructed endpoint component equals the lifted endpoint of  $u$  up to  $O(h^7)$ , and Exp is smooth. Combining with Eq. (24) and the chart norm equivalence gives Eq. (28). The bound Eq. (29) is machine-checked.  $\square$

**Lemma 5** (Endpoint-closure perturbation). *Let  $E(z) = 0$  be the velocity-level/KKT endpoint closure subsystem corrected by  $\mathcal{C}_h$ , and let  $z_u = \Psi_h^G$  be the unclosed endpoint. Under P2 the right-inverse Newton iteration  $z_{l+1} = z_l - B_* E(z_l)$ ,  $z_0 = z_u$ , converges to a closed endpoint  $z_E = \mathcal{C}_h(z_u)$  with  $E(z_E) = 0$ , and any closed endpoint of the form  $z_u + B_* w$  in  $B_{r_E}(z_*)$  satisfies*

$$\|z_E - z_u\| \leq 2M_{E,\text{ri}} \|E(z_u)\| \leq C_E h^7, \quad C_E = 2M_{E,\text{ri}} C_{E,\text{raw}}. \quad (30)$$

*Proof.* Write  $G(w) = E(z_u + B_* w)$ . Since  $DE(z_*)B_* = I$  and  $\|DE(z) - DE(z_*)\| \leq L_E \|z - z_*\| \leq L_E r_E$  on the ball, the mean value inequality gives  $\|G(w_1) - G(w_2) - (w_1 - w_2)\| \leq M_{E,\text{ri}} L_E r_E \|w_1 - w_2\| \leq \frac{1}{2} \|w_1 - w_2\|$  whenever both arguments stay in the ball. The map  $w \mapsto w - G(w)$  is therefore a  $\frac{1}{2}$ -contraction of the ball of radius  $2\|G(0)\| = 2\|E(z_u)\|$  into itself, which stays inside  $B_{r_E}(z_*)$  for  $h \leq h_0$ ; its fixed point  $w_*$  gives  $z_E = z_u + B_* w_*$  with  $E(z_E) = 0$ . For any root of this form, the same inequality with  $w_2 = 0$  yields  $\|w\| \leq 2\|E(z_u)\|$ , hence  $\|z_E - z_u\| = \|B_* w\| \leq 2M_{E,\text{ri}} \|E(z_u)\|$ . The raw-defect hypothesis of P2 completes Eq. (30).  $\square$

The raw endpoint defect  $\|E(z_u)\| \leq C_{E,\text{raw}}h^7$  is a retained P2 hypothesis about the closure functional actually corrected by  $\mathcal{C}_h$ . The endpoint-closure diagnostic in Table 7 records its finite-window scale; it does not discharge the hypothesis.

**Lemma 6** (Inexact Newton tolerance scale). *Under P2 and P6, the stage and output perturbations satisfy Eq. (31):*

$$\begin{aligned} \|\tilde{Z}_A - Z_G\| &\leq 2M\eta_h^{\text{tube}}, \\ \|\mathcal{P}_h(\tilde{Z}_A) - \mathcal{P}_h(Z_G)\| &\leq C_N\eta_h^{\text{tube}} \leq C_Nc_\eta h^7, \quad C_N = 2MM_N. \end{aligned} \quad (31)$$

*Proof.* With Eq. (25) and the mean value inequality of Lemma 2,  $\|\tilde{Z}_A - Z_G\| \leq M(\|F_{A,h}(\tilde{Z}_A)\| + \frac{L_\rho}{2}\|\tilde{Z}_A - Z_G\|)$ , so  $\|\tilde{Z}_A - Z_G\| \leq 2M\eta_h^{\text{tube}}$ . The output bound follows from  $\|D_Z\mathcal{P}_h\| \leq M_N$  on the ball and the envelope  $\eta_h^{\text{tube}} \leq c_\eta h^7$ .  $\square$

The residual inequality and the branch-ball membership in P6 are separate inputs: a small residual alone does not place the iterate on the Gauss-predictor branch, and a converged fixed-tolerance run outside the  $h^7$  envelope is finite-run evidence only.

**Lemma 7** (Newton reaches the solver envelope). *Under P1 and P2 there is  $h_0 > 0$  such that for  $0 < h \leq h_0$  the lifted endpoint predictor  $Z_*^{(0)} = (\mathcal{L}(x_n), \mathcal{L}(x_n), \mathcal{L}(x_n))$  lies in  $B_{\rho/2}(Z_G)$ ; for every predictor  $Z^{(0)} \in B_{\rho/2}(Z_G)$  the simplified Newton iteration  $Z^{(k+1)} = Z^{(k)} - J_h^{-1}F_{A,h}(Z^{(k)})$  stays in  $B_\rho(Z_G)$ , and*

$$\|Z^{(k)} - Z_G\| \leq 2^{-k}\|Z^{(0)} - Z_G\|, \quad \|F_{A,h}(Z^{(k)})\| \leq L_F 2^{-k}\|Z^{(0)} - Z_G\|, \quad (32)$$

where  $L_F$  is a Lipschitz constant of  $F_{A,h}$  on the ball. Consequently the stopping rule  $\|F_{A,h}(Z^{(k)})\| \leq c_\eta h^7$  is met after  $k = O(\log(1/h))$  iterations, and the accepted map of Definition 4 is well defined on the whole tube for small steps whenever the stage solve starts in  $B_{\rho/2}(Z_G)$ : with the lifted endpoint predictor, P6 is achievable by construction.

*Proof.* The stages of  $Z_G$  are  $\mathcal{L}(Y_k)$  with  $\|Y_k - x_n\| = O(h)$  on the compact tube, so  $\|Z_*^{(0)} - Z_G\| = O(h)$  and  $Z_*^{(0)} \in B_{\rho/2}(Z_G)$  for small  $h$ . The linearization inequality of P2 makes the simplified Newton map a  $\frac{1}{2}$ -contraction of  $B_\rho(Z_G)$  into itself with fixed point  $Z_G$  (Lemma 2), which gives the first bound

in Eq. (32); the second follows from  $F_{A,h}(Z_G) = 0$  and the Lipschitz bound. The implementation recomputes the Jacobian at every iterate; for that full Newton iteration P2 gives  $\|D_Z F_{A,h}(Z) - J_h\| \leq L\|Z - Z_G\| \leq L\rho$  on the ball, so the perturbation bound of Lemma 3 yields  $\|D_Z F_{A,h}(Z)^{-1}\| \leq 2M$ , and the standard estimate  $\|Z^{(k+1)} - Z_G\| \leq ML\|Z^{(k)} - Z_G\|^2 \leq \frac{1}{2}\|Z^{(k)} - Z_G\|$  gives the same decay Eq. (32). The simplified-Newton decay estimate is machine-checked (`simplified_newton_residual_decay`).  $\square$

The predictor of Algorithm 1 is not  $Z_*^{(0)}$ : it extrapolates positions and velocities with a constant acceleration but sets the angular-velocity and angular-acceleration guesses to zero and uses force-balance multiplier estimates, so its distance to  $Z_G$  is of the order of  $\|\omega_n\| + \|\alpha_n\| + \|\lambda_n\|$  rather than  $O(h)$ , and Table 12 shows that the first full Newton step from it does not contract in the Euclidean norm. Lemma 7 therefore covers the implemented run only through the branch-ball membership retained in P6; on the reported grids that membership is confirmed a posteriori by Table 11, whose converged roots are the Gauss roots.

The  $h^7$ -scaled stopping rule is not only achievable in principle; it has been exercised on the accepted smooth horizon  $T = 0.08$  with  $c_\eta = 10^4$  at  $h \in \{0.04, 0.02, 0.01, 0.005\}$ : every step converged with residual at most  $c_\eta h^7$  (the largest recorded ratio of residual to  $h^7$  was  $2.1 \times 10^2$ ), and the finite-window position/velocity slopes were 6.946/6.608, identical to those of a fixed  $10^{-10}$  tolerance on the same window and, like the smooth-sweep slopes of Section 8, above six. That record exercises the rule on the reported grids; it is finite-run evidence and does not discharge P6, which the theorem keeps as a hypothesis because the accepted map is defined by the rule, not by a particular run.

#### 7.4. Local-to-global transfer

**Lemma 8** (Local-to-global transfer on the accepted tube). *Let  $\Psi_h$  be a one-step map on the reduced chart,  $t_n = nh$ , and  $y_{n+1} = \Psi_h(y_n)$  with  $y_0$  the exact initial state. Suppose that for  $0 \leq n < N$  with  $Nh \leq T$  the exact states  $Y_n = \varphi_{t_n}(y_0)$  satisfy  $\|\Psi_h(Y_n) - Y_{n+1}\| \leq C_\ell h^{p+1}$ , that  $\|\Psi_h(y_n) - \Psi_h(Y_n)\| \leq (1 + C_s h)\|y_n - Y_n\|$  whenever  $y_n \in K$ , and that  $C_\ell \Gamma_s(T)h^p \leq d_K$ , where  $\Gamma_s(T)$  is the Gronwall factor of Eq. (33),*

$$\Gamma_s(T) = \begin{cases} \frac{e^{C_s T} - 1}{C_s}, & C_s > 0, \\ T, & C_s = 0. \end{cases} \quad (33)$$

Then  $y_n \in K$  for all  $n \leq N$  and the grid bound Eq. (34) holds,

$$\max_{0 \leq n \leq N} \|y_n - Y_n\| \leq C_\ell \Gamma_s(T) h^p. \quad (34)$$

If moreover the reporting map  $\mathcal{R}$  (chart to reported position and velocity) is  $C_{\mathcal{R}}$ -Lipschitz on  $K$ , then  $\max_n \|\mathcal{R}(y_n) - \mathcal{R}(Y_n)\| \leq C_{\mathcal{R}} C_\ell \Gamma_s(T) h^p$ .

*Proof.* Let  $e_n = \|y_n - Y_n\|$ . By induction, as long as  $y_n \in K$ ,  $e_{n+1} \leq (1 + C_s h) e_n + C_\ell h^{p+1}$ , hence  $e_n \leq C_\ell h^{p+1} \sum_{k < n} (1 + C_s h)^k$ . For  $C_s > 0$  the geometric sum equals  $((1 + C_s h)^n - 1)/(C_s h) \leq (e^{C_s n h} - 1)/(C_s h) \leq \Gamma_s(T)/h$ , and for  $C_s = 0$  it equals  $n \leq T/h$ . Thus  $e_n \leq C_\ell \Gamma_s(T) h^p \leq d_K$ , which keeps  $y_n$  in the tube and closes the induction (first-exit bootstrap). The reporting bound is the Lipschitz property. The lemma is machine-checked.  $\square$

**Definition 4** (Accepted branch-selected one-step map). *The accepted map  $\Psi_h^{\text{G6FVA}}$  is defined on the set  $\mathcal{D}_h \subset K$  of reduced states for which the transition data satisfy P2 and P6: the stage solve is initialized from the Gauss predictor, the computed iterate lies in  $B_\rho(Z_G)$  with proof-norm residual at most  $\eta_h^{\text{tube}}$ , the endpoint is reconstructed by Eqs. (15)–(16) and closed by  $\mathcal{C}_h$ , and the output is  $\Psi_h^{\text{G6FVA}}(y) = \mathcal{P}_h(\tilde{Z}_A)$ . The map is undefined for theorem purposes on transitions violating these conditions; the theorem makes no claim about such steps.*

### 7.5. Main theorem

Theorem 1 is therefore a statement about the accepted branch-selected map on  $\mathcal{D}_h$  and about nothing else: it does not assign an order to transitions outside  $\mathcal{D}_h$ , to arbitrary Newton roots, or to fixed-tolerance runs that leave the  $h^7$  envelope, and it carries no residual-to-error content for mechanism-coverage rows. For those rows, small residual norms alone are not a trajectory-error theorem; a residual-to-error implication would have to prove a bound with a uniform stability or inf-sup constant on the reported branch, and no residual-to-trajectory-error transfer theorem is accepted in this manuscript. Closed-loop residual rows remain diagnostics even when their measured norms are small, and they are not accepted dynamic order rows. Finite solver probes and fixed-tolerance runs may corroborate the accepted branch. Those probes are finite solver-policy diagnostics only; they are not proof inputs and not a substitute for the explicit  $\eta_h^{\text{tube}} \leq c_\eta h^7$  theorem condition.

**Theorem 1** (Conditional sixth-order error theorem for the accepted branch-selected Gauss6/FullVA map). *Let Assumption 1 hold with the tube  $K$ , the proof norms, and the constants fixed before any numerical evidence is read, and let  $C_{\text{loc}} = C_G + C_E + C_N c_\eta$  with  $C_G, C_E, C_N$  from Lemmas 4, 5, and 6. Then there is  $h_0 > 0$  such that for  $0 < h \leq h_0$  and every  $y \in \mathcal{D}_h$ ,*

$$\|\Psi_h^{\text{G6FVA}}(y) - \varphi_h(y)\| \leq C_{\text{loc}} h^7. \quad (35)$$

*Consequently, for every reported grid  $t_n = nh$ ,  $0 \leq n \leq N_h$ ,  $N_h h \leq T$ , whose transitions lie in  $\mathcal{D}_h$  and whose numerical trajectory  $y_{n+1} = \Psi_h^{\text{G6FVA}}(y_n)$  starts from the exact reduced initial state,*

$$\max_{0 \leq n \leq N_h} \|y_n - y(t_n)\| \leq C_{\text{red}} h^6, \quad C_{\text{red}} = C_{\text{loc}} \Gamma_s(T), \quad (36)$$

*and the reported position-velocity map  $\mathcal{R}(y) = (q(y), v(y))$  satisfies*

$$\max_{0 \leq n \leq N_h} \|\mathcal{R}(y_n) - \mathcal{R}(y(t_n))\| \leq C_{qv} h^6, \quad C_{qv} = C_{\mathcal{R}} C_{\text{red}}. \quad (37)$$

*The constants depend on the tube,  $T$ , and the constants of Assumption 1, including  $c_\eta$ , but not on  $h$ ,  $N_h$ , the linear-algebra backend, or the fixed production tolerances of finite diagnostic runs.*

*Proof.* Fix  $y \in \mathcal{D}_h$  and write  $\Psi_h^G = \mathcal{E}_h(Z_G)$ ,  $\Psi_h^E = \mathcal{P}_h(Z_G)$ , and  $\Psi_h^{\text{G6FVA}}(y) = \mathcal{P}_h(\tilde{Z}_A)$ . By Lemma 2 the exact accepted stage root is  $Z_G$ , so there is no stage-root perturbation term, and the three-term chain Eq. (38) gives

$$\begin{aligned} \|\Psi_h^{\text{G6FVA}}(y) - \varphi_h(y)\| &\leq \|\Psi_h^G(y) - \varphi_h(y)\| + \|\Psi_h^E(y) - \Psi_h^G(y)\| \\ &\quad + \|\Psi_h^{\text{G6FVA}}(y) - \Psi_h^E(y)\| \\ &\leq C_G h^7 + C_E h^7 + C_N c_\eta h^7 = C_{\text{loc}} h^7 \end{aligned} \quad (38)$$

by Lemmas 4, 5, and 6, after shrinking  $h_0$  so that all three lemmas apply on the tube. This is Eq. (35). Choose  $h_0$  additionally so that  $C_{\text{loc}} \Gamma_s(T) h_0^6 \leq d_K$ . Lemma 8 with  $p = 6$ ,  $C_\ell = C_{\text{loc}}$ , the local defect Eq. (35) at the exact grid states, and the stability scale Eq. (23) gives Eq. (36); the reporting-map clause gives Eq. (37). The constants  $C_G, C_E, C_N, c_\eta, C_s, C_{\mathcal{R}}$  are fixed by Assumption 1 before any reported grid is considered, so  $C_{\text{loc}}$ ,  $C_{\text{red}}$ , and  $C_{qv}$  are independent of  $h$  and  $N_h$ .  $\square$

*Scope and non-claims of Theorem 1.* The theorem is a same-initial-state reduced-chart and reported position-velocity estimate on the reported grid for transitions in  $\mathcal{D}_h$ . It does not claim an unconditional order: it does not close P6 and supplies no fixed-tolerance asymptotic proof of the solver envelope, and it performs no P7 residual-to-error promotion. It does not prove residual, reaction, multiplier, or constraint-output error estimates; it does not prove a residual-to-error transfer for mechanism-coverage rows (the P7 boundary below); it does not turn fixed-tolerance finite runs into an asymptotic solver-policy proof (P6 is retained); and it does not supply source-paper reproduction. If  $N_h h < T$  the estimate is a prefix-grid estimate and says nothing about the leftover interval. The observed smooth slopes 7.161/7.066 in Section 8 are consistency diagnostics of the accepted branch, not proof inputs. Because the stage residual vanishes identically, the method’s order is inherited from the reduced Gauss collocation; the absolute-coordinate formulation contributes the exact three-level constraint enforcement, the multipliers and reaction forces, the friction law through the multipliers, and the Lie-group endpoint reconstruction, but no additional stage-level approximation.

#### 7.6. Formal-order comparison and output boundary

**Lemma 9** (Order of the local paper-style TFE target). *The proof-level order target for the  $m = 3$  Gauss–Lobatto paper-style TFE formula used in this manuscript is five.*

*Proof.* The source-paper family is compared through the temporal finite-element formula-order convention  $p_{\text{TFE}}(m) = 2m - 1$  in Eq. (19). Substituting  $m = 3$  gives  $p_{\text{TFE}}(3) = 5$ . This is the formula-order comparator used here. It is distinct from the source paper’s reported DAE pendulum observation, where order reduction can make the implemented  $m = 3$  run appear fourth order.  $\square$

**Proposition 1** (Conditional order comparison). *Under Assumption 1 and the current reproducibility contract, Gauss6/FullVA has higher formal order than the local paper-style  $m = 3$  Gauss–Lobatto TFE formula target in the narrow order-comparison sense.*

*Proof.* Theorem 1, built on the exact stage identity of Lemma 1, gives the conditional route to the sixth-order method claim for Gauss6/FullVA. Lemma 9 gives the fifth-order local paper-style  $m = 3$  comparator. This

theorem–lemma pair is the formal-order proof of the proposition. The generated h-sweep records observed smooth position and velocity orders 7.161 and 7.066 only as consistency evidence after that comparison; those finite-window slopes are not proof inputs. The four-example validation criterion is a mechanism-coverage criterion, not a single order theorem: the single- and double-pendulum dynamic FullVA rows provide dynamic order evidence, while the four-link and slider-crank rows are driven closed-loop kinematic/reaction checks whose trajectory differences sit at the roundoff floor and are not interpreted as method-order estimates. Therefore the conditional formal-order comparison is supported only inside the smooth accepted dynamic path specified in Assumption 1 and Theorem 1. The comparison inherits the compact-tube branch selection and solver-scale restrictions of Theorem 1: Newton is initialized from the Gauss predictor on the accepted branch, and the nonlinear residual budget is the theorem-level  $\eta_h^{\text{tube}} \leq c_\eta h^7$  policy. It is not a theorem about arbitrary Newton initializations, remote roots, fixed production tolerances, or global solver convergence. The proposition does not compare error constants, wall-clock performance against independently implemented baselines, robustness in the nonsmooth sharp-friction regime, four-link/slider-crank dynamic convergence order, or complete source-paper residual reproduction. Those exclusions define the claim boundary; in particular, the method is not a complete source-paper temporal finite-element residual replacement.  $\square$

**Corollary 1** (Non-claim). *Proposition 1 does not prove that the implemented nonlinear residual is the complete source-paper temporal finite-element residual.*

*Proof.* The accepted method solves Eq. (6). A complete source-paper residual reproduction would require replacing every stage row in that solve by accepted paper-derived weak rows and passing the same validation criteria. That stronger criterion remains open.  $\square$

**Definition 5** (P7 boundary-certificate template). *For a mechanism-coverage row family on a fixed accepted branch, an accepted P7 residual-to-error certificate would be a same-branch tuple*

$$\mathcal{T}_{\text{P7}} = (\mathcal{R}_h^{\text{mech}}, \|\cdot\|_{\text{traj}}, C_{\text{P7}}, \rho_{\text{res}}, \varepsilon_{\text{floor}}, \varepsilon_{\text{est}}, \mathcal{G}_{\text{dyn}}). \quad (39)$$

*and this tuple is fixed before the residual table is inspected, with the following properties. Equation (39) fixes the object whose transfer bound is Eq. (40).*

First,  $\mathcal{R}_h^{\text{mech}}$  is the residual operator actually reported for the mechanism rows, with the same branch, time grid, row scaling, and norm used in the table. Second,  $\|\cdot\|_{\text{traj}}$  is the trajectory error norm in which a dynamic-order row would be claimed. Third,  $C_{\text{P7}}$  is independent of  $h$  on the same compact branch. Fourth,  $\rho_{\text{res}}(h)$  is an a priori residual-consistency rate for inserting the exact branch into the mechanism residual rows. Fifth,  $\varepsilon_{\text{floor}}(h)$  and  $\varepsilon_{\text{est}}(h)$  bound, respectively, reference-floor and estimator/calibration errors on the same reported campaign. Sixth,  $\mathcal{G}_{\text{dyn}}$  supplies the dynamic identity, closed-loop DAE stability or inf-sup estimate, and output binding needed to connect residual rows to trajectory variables. Seventh, the dynamic-order campaign has a branch-compatible refinement set on which all of these constants and rates are valid. The stability or inf-sup clause is a nullspace exclusion. It must rule out same-branch perturbations that are invisible to the reported mechanism residual but visible in the trajectory norm; otherwise a constraint or reaction residual can be small while the reported position-velocity error is not controlled. Thus the residual operator, row weights, gauge conventions, and trajectory output variables must be fixed together before any residual row is promoted.

The certificate would have to imply Eq. (40), uniformly for all sufficiently small reported step sizes in that campaign,

$$\|e_h\|_{\text{traj}} \leq C_{\text{P7}} \left( \|\mathcal{R}_h^{\text{mech}}\|_{\text{res}} + \rho_{\text{res}}(h) + \varepsilon_{\text{floor}}(h) + \varepsilon_{\text{est}}(h) \right). \quad (40)$$

Definition 5 is a closure criterion, not a theorem premise of Theorem 1: no such tuple is proved or invoked in this manuscript.

**Corollary 2** (Residual-to-error transfer boundary). *The four-link and slider-crank residual/reaction rows in the validation section, and the common-reference or source-policy residual rows reported elsewhere, do not become accepted dynamic-order or trajectory-error rows in this manuscript.*

*Proof.* This exclusion is not a residual-transfer theorem. Those rows are mechanism-coverage evidence for the accepted FullVA residual vocabulary, not residual-to-error evidence for the reported trajectory norm. Promoting them to trajectory-error/order evidence would require seven residual-to-error obligations to close on the same branch: dynamic residual identity, residual-consistency rate, closed-loop DAE stability or inf-sup control, calibrated estimator, reference-floor exclusion, coarse-first dynamic-order campaign, and

a manuscript transfer theorem. Equivalently, the closure would have to instantiate the same-branch tuple in Definition 5. Since that joint criterion remains open, the rows remain diagnostics. Equivalently, that separate promotion would require an additional same-branch residual operator with a uniform inverse, fixed independently of the observed residual table. More formally, a proof of such a transfer would first have to define a same-branch residual transfer operator on the reported closed-loop branch, a trajectory norm  $\|e_h\|_{\text{traj}}$  matching the claimed order rows, and an  $h$ -independent constant  $C_{\text{P7}}$  such that, for all sufficiently small  $h$  in the claimed dynamic-order campaign, the transfer bound in Eq. (41) would have to hold:

$$\|e_h\|_{\text{traj}} \leq C_{\text{P7}} \left( \|\mathcal{R}_h^{\text{mech}}\|_{\text{res}} + \rho_{\text{res}}(h) + \varepsilon_{\text{floor}}(h) + \varepsilon_{\text{est}}(h) \right), \quad (41)$$

with the residual identity, residual-consistency rate, floor exclusion, and estimator terms all attached to the same branch. The present residual and reaction tables provide none of this transfer operator: they report small mechanism residuals and consistent reactions, but they do not provide the uniform inverse/stability constant or the reference-floor and estimator closure needed to convert those residuals into trajectory-error order. In particular, they do not exclude residual-null directions on the reported closed-loop branch. A perturbation that changes the reported trajectory while leaving the chosen residual/reaction rows unchanged would violate any uniform P7 transfer estimate, so finite residual smallness is not enough to define the missing inverse. The manuscript retains the scope rule that these rows are not promoted; this retained boundary does not supply any stability, estimator, floor-exclusion, campaign, or manuscript-theorem component of the open P7 route.  $\square$

### 7.7. External comparison boundary

External source-paper comparison is kept outside the main method and proof claims. The accepted evidence below supports the branch-selected Gauss6/FullVA method, the conditional sixth-order theorem, and method-side mechanism coverage. The TFE and public-code source-policy rows are retained only as diagnostic boundary records; they close no source-paper superiority row and no theorem input. The detailed same-test promotion criterion, common-reference tables, and source-policy runner gaps are recorded in Section Appendix A so that the main numerical section is read in the intended evidence order: dynamic-order evidence first, mechanism-coverage evidence second, external-comparison diagnostics last.

Table 8: Proof-dependency map for Proposition 1.

| Theorem component           | Mathematical input                                                                                            | Role in the claim                                                                        |
|-----------------------------|---------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Accepted one-step map       | Stage system Eq. (11), endpoint reconstruction, and endpoint closure in Sections 5–6.                         | Definition fixed for the theorem and numerical protocol.                                 |
| Exact stage identity        | Lemma 1: the lifted reduced Gauss stage satisfies all 132 rows; the stage solve is reduced Gauss collocation. | Proved; removes the stage-residual perturbation term.                                    |
| Gauss endpoint defect       | Lemma 4: classical Gauss local error plus the Gauss-weight quadrature defect Eq. (29).                        | $C_G h^7$ ; cited collocation theory plus a machine-checked quadrature bound.            |
| Endpoint closure            | Lemma 5: right inverse, closed anchor, raw defect hypothesis of P2.                                           | $C_E h^7$ ; retained P2 hypothesis for the raw defect.                                   |
| Inexact Newton              | Lemma 6 under the P6 envelope $\eta_h^{\text{tube}} \leq c_\eta h^7$ .                                        | $C_N c_\eta h^7$ ; fixed production tolerances do not close P6.                          |
| P2 from P1                  | Lemma 3: block-triangular $h \rightarrow 0$ Jacobian, Neumann bound Eq. (27).                                 | P2 from P1 for $h \leq h_0$ ; P2 kept as stated interface.                               |
| Solver envelope             | Lemma 7: $2^{-k}$ Newton decay Eq. (32) from the ball.                                                        | $c_\eta h^7$ rule reachable from the ball; P6 keeps the predictor’s ball membership.     |
| Local-to-global transfer    | Lemma 8: $1 + C_s h$ stability, tube retention, $\Gamma_s(T)$ .                                               | Grid-point transfer; constants independent of $h, N_h$ .                                 |
| Comparator order            | Lemma 9: local $m = 3$ expected order $2m - 1 = 5$ .                                                          | Formal-order comparator only.                                                            |
| Four examples               | Section 9.                                                                                                    | Mechanism coverage; closed-loop rows are residual/reaction checks, not order-proof rows. |
| Residual-to-error promotion | No transfer theorem is accepted for residual-only rows (Definition 5).                                        | Excluded output boundary (P7).                                                           |
| Non-claims                  | Complete source-paper residual reproduction and full-TFE stage replacement.                                   | Excluded stronger criterion.                                                             |

## 8. Numerical evidence

The numerical section should be read as a sequence of evidence blocks rather than as a single undifferentiated benchmark table. First, the smooth cylindrical-chain sweep tests the accepted 132-row Gauss6/FullVA stage solve and is finite-run consistency evidence after the conditional order-six theorem; it is not a premise that discharges the compact-tube, endpoint, or solver-scale hypotheses. Second, the four ASME-style mechanisms test the lower-pair FullVA vocabulary across driven, open-chain, and closed-loop settings; only the single and double pendula are used as accepted dynamic-order examples, while the closed-loop examples supply constraint, reaction, and coarse-window trajectory diagnostics only. Third, the source-paper and public-code rows are comparison diagnostics whose role is to expose what remains open in a fair same-test campaign. Finally, the work and solver tables make the evidence reproducible by recording Newton counts, tolerances, and runtime, but they do not by themselves turn the diagnostic baseline rows into superiority claims.

The recorded h-sweep uses  $h = \{0.04, 0.02, 0.01\}$  and an  $h = 0.005$  reference for the smooth cylindrical-chain benchmark. Table 9 gives the headline order evidence. Reference-solution boundary: every observed order reported from a finite  $h_{\text{ref}}$  row is a finite-reference consistency diagnostic, not a replacement for the exact-flow comparison used in Theorem 1. The reference step, output norm, and reported grid are fixed before the fit is computed. A finite reference row cannot discharge the compact-tube regularity, endpoint-closure, P6 solver-scale, or P7 residual-to-error boundaries, cannot be used to tune the reported norm after the fact, and cannot exclude a reference floor without a separate refinement or analytic-reference argument.

Table 9: Observed orders for the accepted Gauss6/FullVA path.

| Case                         | Step sizes       | Position order | Velocity order |
|------------------------------|------------------|----------------|----------------|
| Smooth cylindrical chain     | 0.04, 0.02, 0.01 | 7.161          | 7.066          |
| Sharp friction, coarse sweep | 0.04, 0.02, 0.01 | 1.894          | 2.683          |

Table 10: Smooth cylindrical-chain projected-endpoint errors.

| $h$  | Steps | Position error          | Velocity error          | Newton iterations |
|------|-------|-------------------------|-------------------------|-------------------|
| 0.04 | 2     | $1.761 \times 10^{-7}$  | $1.007 \times 10^{-5}$  | 12                |
| 0.02 | 4     | $1.838 \times 10^{-9}$  | $2.954 \times 10^{-8}$  | 24                |
| 0.01 | 8     | $8.602 \times 10^{-12}$ | $5.609 \times 10^{-10}$ | 48                |

The slopes 7.161 and 7.066 in Tables 9–10 should not be read as a seventh-order method claim. The theorem above claims sixth order under an  $O(h^7)$  local-defect contract. The larger fitted slopes come from a short, smooth, projected-endpoint window in which the leading sixth-order error coefficient is small relative to higher-order terms, while the finest row is already close to reference, nonlinear-tolerance, and floating-point floors. Consequently the paper treats these values as post-theorem consistency evidence for the order-six behavior on the accepted smooth path, not as a proof premise or a new asymptotic order. A wider asymptotic sweep with a compact-tube  $\eta_h^{\text{tube}} \leq c_\eta h^7$  nonlinear tolerance would be needed before claiming more than order six, and the closed-loop four-link and slider-crank rows are still not promoted to dynamic order estimates. Under the exact stage identity the method is reduced Gauss collocation, so slopes above six on a short window are the expected pre-asymptotic behaviour of a sixth-order collocation method, not a signal of a different order.

*Exact stage identity diagnostic.* Lemma 1 has a direct numerical counterpart. Each converged stage vector of the smooth  $h$ -sweep, obtained with a tight Newton tolerance of  $10^{-13}$  in the proof norm, is read in the reduced joint-coordinate chart, and the reduced Gauss collocation defects  $\Delta_i[s_j]$ ,  $\Delta_i[\dot{s}_j]$ ,  $\Delta_i[\theta_j]$ ,  $\Delta_i[\dot{\theta}_j]$  of Eq. (10) are evaluated together with the component of  $d_j, \dot{d}_j, \ddot{d}_j$  orthogonal to  $n_j$  and the factorization of the implemented sliding row. Table 11 shows all of them at roundoff over the accepted horizon  $T = 0.08$ , with six Newton iterations per step at every step size. This is a consistency check of the transcription used in the lemma, not a proof input.

*Constants of the P2 reduction.* Lemmas 3 and 7 have a numerical counterpart as well. On the same solves the implemented Jacobian is evaluated at the converged stage vector, giving  $J_h$ , and with  $h = 0$  at the root of the

Table 11: Exact stage identity diagnostic on the smooth cylindrical chain.

| $h$  | Steps | Newton<br>iters. | Max stage<br>residual | Max reduced<br>defect | Max $\perp n_j$<br>component |
|------|-------|------------------|-----------------------|-----------------------|------------------------------|
| 0.04 | 2     | 12               | $6.8 \times 10^{-14}$ | $9.0 \times 10^{-15}$ | $1.5 \times 10^{-15}$        |
| 0.02 | 4     | 24               | $1.5 \times 10^{-14}$ | $5.5 \times 10^{-16}$ | $1.9 \times 10^{-15}$        |
| 0.01 | 8     | 48               | $1.6 \times 10^{-14}$ | $5.8 \times 10^{-16}$ | $2.0 \times 10^{-15}$        |

$h = 0$  stage system, which reproduces the lifted endpoint state to  $10^{-12}$ , giving  $J_0$ . Table 12 records, in the Euclidean norm on the implemented stage layout,  $M_0 = \max_n \|J_0^{-1}\|$ ,  $C_J = \max_n \|J_h - J_0\|/h$ , the Neumann margin  $M_0\|J_h - J_0\|$ , the observed ratio  $\|J_h^{-1}\|/\|J_0^{-1}\|$ , the threshold  $h_0 = 1/(2M_0C_J)$  implied by the Neumann argument, and the distances to  $Z_G$  of the lifted endpoint predictor and of the Algorithm 1 predictor. The conclusion of Lemma 3 holds on every grid with ratio at most 1.55, but its sufficient condition  $M_0\|J_h - J_0\| \leq \frac{1}{2}$  would require  $h \leq 3 \times 10^{-5}$ : the Neumann threshold is pessimistic by three orders of magnitude in this norm. The lifted predictor distance decreases with  $h$  as the lemma predicts, whereas the Algorithm 1 predictor stays at distance about 50 because its angular-velocity and angular-acceleration guesses are zero, and the first full Newton step from it increases the distance by up to 18% before the iteration converges. These are diagnostics of the lemma constants, not proof inputs.

Table 12: Constants of Lemmas 3 and 7 on the smooth cylindrical chain, Euclidean norm on the implemented stage layout:  $M_0 = \max_n \|J_0^{-1}\|$ ,  $C_J = \max_n \|J_h - J_0\|/h$ , Neumann margin  $M_0\|J_h - J_0\|$ , inverse ratio  $\|J_h^{-1}\|/\|J_0^{-1}\|$ , implied  $h_0 = 1/(2M_0C_J)$ , distances of the lifted endpoint and Algorithm 1 predictors to  $Z_G$ , and the ratio  $\|Z^{(1)} - Z_G\|/\|Z^{(0)} - Z_G\|$  of the first full Newton step from the Algorithm 1 predictor.

| $h$  | $M_0$ | $C_J$             | Neumann<br>margin | inverse<br>ratio | implied $h_0$        | lifted<br>pred. | Alg. 1<br>pred. | first-<br>step<br>ratio |
|------|-------|-------------------|-------------------|------------------|----------------------|-----------------|-----------------|-------------------------|
| 0.04 | 18.2  | $7.5 \times 10^2$ | $5.4 \times 10^2$ | 1.55             | $3.7 \times 10^{-5}$ | 3.23            | 50.2            | 1.12                    |
| 0.02 | 24.3  | $8.2 \times 10^2$ | $3.0 \times 10^2$ | 1.26             | $3.3 \times 10^{-5}$ | 2.11            | 50.3            | 1.16                    |
| 0.01 | 28.4  | $9.6 \times 10^2$ | $2.0 \times 10^2$ | 1.18             | $2.5 \times 10^{-5}$ | 1.07            | 50.4            | 1.18                    |

Table 14 is retained only as a non-promotion record for the current source-paper comparator status. The paper-style column is an implemented all-row TFE diagnostic using the terminal Lobatto-node output, a consistent initial

Table 13: Order-estimate acceptance policy used in the numerical section.

| Evidence row                    | Step/reference policy                                                        | Order information                                                                                                                       | Manuscript role                                                                                                                 |
|---------------------------------|------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Smooth cylindrical chain        | $h = 0.04, 0.02, 0.01$ against $h = 0.005$ .                                 | Position/velocity slopes 7.161/7.066.                                                                                                   | Post-theorem finite-run consistency check for the conditional order-six theorem; not a proof premise or seventh-order claim.    |
| Single pendulum                 | Accepted dynamic FullVA rows plus coarse public-window rows.                 | Minimum accepted dynamic order above six; coarse local position order about 6.05.                                                       | Accepted dynamic-order example and coarse-first comparison row.                                                                 |
| Double pendulum                 | Method-side FullVA dynamic rows plus coarse public-window rows.              | Minimum accepted method-side order above six; coarse public-window position/velocity slopes 7.951/7.042.                                | Accepted dynamic-order example and coarse-first comparison row.                                                                 |
| Four-link and slider-crank      | Closed-loop kinematic/reaction rows and strict common-reference coarse rows. | Local coarse trajectory-diagnostic candidates are near six, but floor-limited endpoint rows are not used as asymptotic order estimates. | Mechanism-coverage and work/precision evidence; not accepted asymptotic dynamic-order proof or an external dynamic-order claim. |
| Paper-style TFE all-row attempt | Same smooth cylindrical-chain steps as the accepted row.                     | Position/velocity slopes 4.046/4.420.                                                                                                   | Internal diagnostic only; not a source-paper same-test row and not an accepted full-TFE replacement.                            |
| Public baseline families        | Public-code and same-window rows where available.                            | Useful order/time rows, including nonconvergence rows for some coarse settings.                                                         | External-baseline evidence in progress; no external-superiority claim.                                                          |

Table 14: Non-promoted same-step diagnostic with a TFE-style all-row attempt.

| $h$  | Accepted Gauss6/FullVA                                                           | Paper-style all-row TFE diagnostic                                              | Diagnostic status |
|------|----------------------------------------------------------------------------------|---------------------------------------------------------------------------------|-------------------|
| 0.04 | Position $1.761 \times 10^{-7}$ ; velocity $1.007 \times 10^{-5}$ ; 12 Newton.   | Position $9.064 \times 10^{-8}$ ; velocity $1.238 \times 10^{-5}$ ; 13 Newton.  | Not accepted.     |
| 0.02 | Position $1.838 \times 10^{-9}$ ; velocity $2.954 \times 10^{-8}$ ; 24 Newton.   | Position $3.378 \times 10^{-9}$ ; velocity $1.931 \times 10^{-7}$ ; 21 Newton.  | Not accepted.     |
| 0.01 | Position $8.602 \times 10^{-12}$ ; velocity $5.609 \times 10^{-10}$ ; 48 Newton. | Position $3.322 \times 10^{-10}$ ; velocity $2.701 \times 10^{-8}$ ; 37 Newton. | Not accepted.     |

$z_0$  policy, and an equilibrated Newton linear solve. Its recorded smooth orders are 4.046 in position and 4.420 in velocity. Because the start policy is not a paper-derived recurrent policy and the full-TFE replacement criterion is not met by the current evidence package, this row is not an accepted fair baseline and is not part of the main order evidence.

Table 15: Work and solver diagnostics for accepted evidence rows.

| Evidence row             | Finest work                         | Runtime (s) | Finest accepted value                                                            |
|--------------------------|-------------------------------------|-------------|----------------------------------------------------------------------------------|
| Smooth cylindrical chain | $h = 0.01$ ; 8 steps / 48 Newton    | 0.199       | Position $8.602 \times 10^{-12}$ ; velocity $5.609 \times 10^{-10}$ .            |
| Single pendulum          | $h = 0.05$ ; 4 steps / 8 Newton     | 0.013       | Position $7.374 \times 10^{-12}$ ; max stage residual $3.664 \times 10^{-12}$ .  |
| Double pendulum          | $h = 0.01$ ; 20 steps / 60 Newton   | 3.374       | Orientation $1.597 \times 10^{-15}$ ; angular velocity $4.867 \times 10^{-15}$ . |
| Four-link closed loop    | $h = 0.005$ ; 40 steps / 164 Newton | 0.118       | Max dynamics residual $1.338 \times 10^{-13}$ .                                  |
| Slider-crank closed loop | $h = 0.005$ ; 40 steps / 163 Newton | 0.122       | Max dynamics residual $6.492 \times 10^{-15}$ .                                  |

Table 16: Nonlinear solver tolerances used for the reported evidence rows.

| Evidence row             | Nonlinear solve                                                                                                                                           | Stopping rule                                                                             | Finest count            |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|-------------------------|
| Cylindrical chain        | 132-row Gauss6/FullVA residual with dense AD Jacobian and dense linear solve; rank diagnostics use tolerance $10^{-10}$ .                                 | Stop when $\ R\ _2 < 10^{-11}$ or $\ \Delta Z\ _2 < 10^{-11}$ ; cap 80 Newton iterations. | 48 Newton.              |
| Single pendulum          | Absolute-coordinate driven FullVA residual with AD Jacobian and dense linear solve.                                                                       | Stop when $\ R\ _2 < 10^{-11}$ or $\ \Delta Z\ _2 < 10^{-12}$ ; cap 32 Newton iterations. | 8 Newton.               |
| Double pendulum          | Imported double-revolute FullVA harness with three Gauss stages, AD Jacobian, and dense linear solve.                                                     | Stop when $\ R\ _2 < 10^{-11}$ or $\ \Delta Z\ _2 < 10^{-11}$ ; cap 64 Newton iterations. | 60 Newton.              |
| Four-link closed loop    | Local fully constrained kinematic FullVA solve for $\Phi(q, t) = 0$ followed by linear solves for $\Phi_q \dot{q} = \nu$ and $\Phi_q \ddot{q} = \gamma$ . | Position Newton stops when $\ \Delta q\ _2 < 10^{-12}$ ; cap 30 Newton iterations.        | 164 Newton; max 4/step. |
| Slider-crank closed loop | Same local closed-loop kinematic FullVA policy as the four-link example.                                                                                  | Position Newton stops when $\ \Delta q\ _2 < 10^{-12}$ ; cap 30 Newton iterations.        | 163 Newton; max 4/step. |

For the asymptotic theorem, Table 16 should be read with the inexact-Newton policy of Lemma 6: a proof-level h-sweep must use the nonlinear residual threshold in Eq. (42),

$$\eta_h^{\text{tube}} \leq c_\eta h^7 \quad (42)$$

on the compact-tube branch, or an equivalent update criterion that implies the same stage-solution error. The fixed absolute tolerances in the recorded runs are retained for reproducibility of the reported runs and explain the reported Newton counts; they are not by themselves a derivation of the asymptotic solver-error bound. A solver-scale check of the existing summary residuals records roundoff-scale linear and stage residuals, but the ratios to  $h^7$  grow at the finest reference scale. A separate short-trajectory scaled-tolerance probe records the finite-log reported-grid maximum  $\eta_h^{\text{reported}} := \max_{0 \leq n < N_h} \eta_{h,n} = \hat{\eta}_h^{\text{log}} \leq c_\eta h^7$  with  $c_\eta = 10^4$  at  $h = 0.04, 0.02, 0.01, 0.005$  through  $T = 0.08$ ; all four diagnostic rows and all 30 trajectory steps pass the finite diagnostic filter, and the maximum final residual divided by  $h^7$  is 210.891. This diagnostic pass is not a theorem-level P6 proof. It is a useful finite P6 diagnostic record, broader than a one-step smoke test, and it records reported-window consistency only rather than solver-policy closure. A companion finite tolerance-regime sweep compares fixed  $10^{-10}$ ,  $ch^7$ , and  $ch^8$  stopping rules against a local  $h = 0.0025$  reference over the same window. The fixed and  $h^7$  policies give position/velocity fits 6.946/6.608, while the  $h^8$  policy gives 6.946/6.608. The two shrinking-target policies are separately recorded as eight finite  $h$ -scaled rows and 60  $h$ -scaled trajectory steps, with position/velocity order floor 6.946/6.608. Thus, on this finite diagnostic window, the fixed-tolerance row and the  $h$ -scaled rows have the same recorded order floor; this is a finite-window tolerance comparison only, not P6 instantiation or a theorem-level solver-policy proof. The scaled policies exercise the theorem's shrinking tolerance form on this finite window, but the sweep is still not a theorem-level scaled-tolerance proof over every reported trajectory and not a scaled-tolerance proof under a uniform all-transition policy for the reported trajectory. The proof therefore remains explicitly conditional on the  $\eta_h^{\text{tube}} \leq c_\eta h^7$  compact-tube policy. The reported-log quantity is only  $\eta_h^{\text{reported}} = \max_{0 \leq n < N_h} \eta_{h,n}$ , also denoted  $\hat{\eta}_h^{\text{log}}$  in diagnostic tables, after the accepted branch, proof norm, row scaling, and  $c_\eta$  have already been fixed; it cannot select the branch, norm, or theorem constant.

Tables 15–16 and Table 14 are solver-scale and diagnostic comparison tables, not accepted fair baseline comparisons. Runtime values are single-machine recorded timings, and the mechanism rows report the accepted finest-step evidence for each validation path. A publication-level comparison still requires accepted implemented baselines and cross-method work/precision curves on the same problems; these tables and Figure 2

only prevent the accepted evidence from being reported without any work, tolerance, Newton-iteration, or source-paper diagnostic context. For the four-link and slider-crank rows, the finest-step residual and reaction entries are read through the P7 boundary stated after Theorem 1: they demonstrate closed-loop mechanism coverage and consistency of the constructed FullVA rows, but they are not trajectory-order rows unless an accepted residual-to-error transfer is proved.

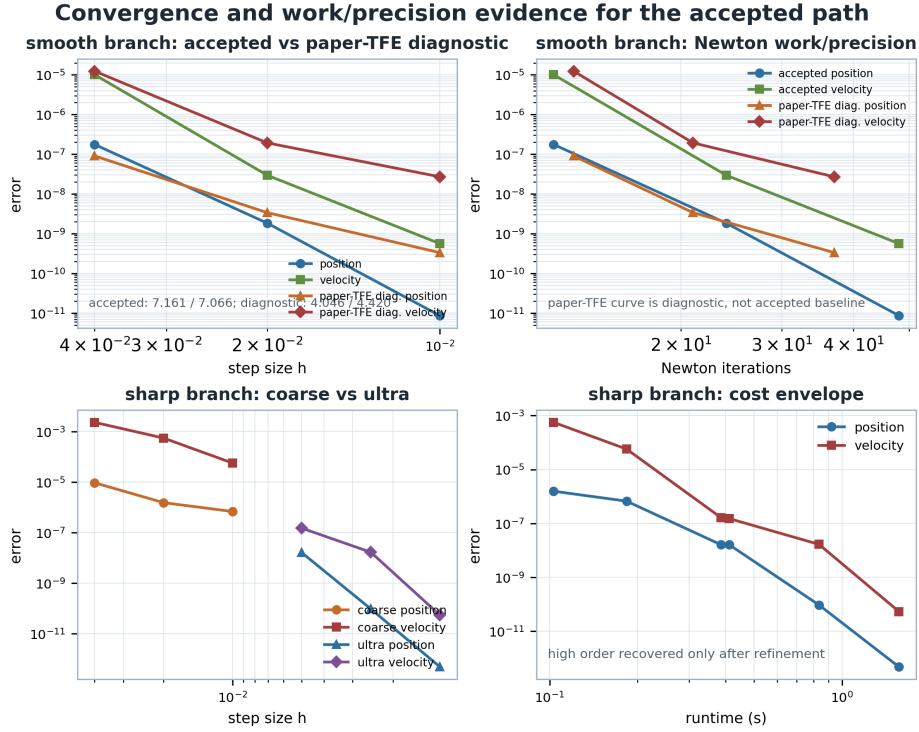


Figure 2: Convergence and work/precision evidence for the accepted Gauss6/FullVA path and the implemented source-paper all-row TFE diagnostic. The top row compares the smooth cylindrical-chain errors and Newton-work/error curves against the diagnostic paper-style all-row attempt; that curve is not an accepted fair baseline. The bottom row contrasts the sharp-friction coarse sweep with the refined cost envelope that recovers high order only at additional runtime.

The sharp-friction coarse sweep is not used as the main order claim. It is reported as a limitation: deeper fixed refinement recovers high order at increased cost, while the practical coarse regime remains order-reduced. The recorded sharp-friction diagnostic shows the practical tradeoff: the coarse

$h = \{0.04, 0.02, 0.01\}$  sweep gives only 1.894/2.683 position/velocity order, while ultra refinement over  $h = \{0.005, 0.0025, 0.00125\}$  against  $h = 0.000625$  recovers 7.521/5.725 all-point position/velocity order. The high-order recovery requires an approximately  $7.25\times$  runtime increase relative to the  $h = 0.01$  baseline, so the paper keeps this as a practical cost caveat rather than a solved production setting.

## 9. Four-example mechanism validation

The method-side evidence covers four ASME-style examples, with different accepted roles. This section states the mechanism content used by the validation rows and keeps their evidentiary roles distinct. The examples are not treated as interchangeable benchmark names: each one exercises a different part of the FullVA residual. These validation rows are downstream evidence for method behavior; they do not feed back into the constants, hypotheses, or residual-certificate slot of Theorem 1. In particular, closed-loop residual and reaction rows remain coverage/diagnostic rows unless the separate P7 residual-to-error boundary is discharged.

### 9.1. Driven single pendulum

The single pendulum is a rigid bar of length 4.0 m, square side length 0.05 m, density 7800 kg/m<sup>3</sup>, and gravity  $g = 9.81$  m/s<sup>2</sup>. The center of mass is 2.0 m from the fixed pivot. The accepted driven trajectory is

$$\theta(t) = \frac{\pi}{2} + \frac{\pi}{4} \cos(2t), \quad (43)$$

Equation (43) defines the accepted driven single-pendulum trajectory, with the corresponding analytic  $\dot{\theta}$  and  $\ddot{\theta}$  used for reference values. The constraint set consists of a fixed pivot, two DP1 revolute-axis constraints, and a DP1 drive constraint. The strongest accepted row is the absolute-coordinate driven FullVA residual: each Gauss stage solves for orientation, center-of-mass position, velocity, acceleration, angular velocity, angular acceleration, pivot reaction force, axis multipliers, and drive multiplier. Its square residual includes pivot velocity and pivot acceleration rows, two revolute-axis rows, and the acceleration-level drive row.

### 9.2. Double pendulum

The double pendulum uses two rigid bars with lengths 4.0 m and 2.0 m, the same square side length, density, and gravity as the single-pendulum case, and two revolute joints. The first body is pinned to ground and the second body is pinned to the first body. Friction and external torques are disabled in the accepted method-side comparison. Because the available legacy rA reference shows an  $h$ -independent discrepancy under the coordinate convention change, the accepted reference policy is a nested local Gauss6/FullVA self-reference at  $h = 0.005$ ; the rA comparison is retained only as a diagnostic reference-floor test.

### 9.3. Four-link closed loop

The four-link example is a driven closed-loop lower-pair graph with three moving rigid bodies and 18 scalar constraints. The constraint inventory is 12 coordinate-difference rows and 6 DP1 rows. Since the graph is fully constrained, the accepted evidence is not a free dynamic convergence claim. At each output time the local FullVA solve enforces Eq. (44),

$$\Phi(q, t) = 0, \quad \Phi_q(q, t)\dot{q} = \nu(q, t), \quad \Phi_q(q, t)\ddot{q} = \gamma(q, \dot{q}, t), \quad (44)$$

then reconstructs multipliers and checks Newton–Euler translational and rotational residuals. The largest recorded dynamics residual is  $1.338 \times 10^{-13}$ .

### 9.4. Slider-crank closed loop

The slider-crank example is also a three-body fully constrained graph with 18 scalar constraints, but it includes prismatic geometry through DP2 and distance rows. Its inventory is 6 coordinate-difference rows, 7 DP1 rows, 4 DP2 rows, and one distance row. The accepted solve follows the same closed-loop FullVA policy as the four-link mechanism. The largest recorded constraint norm is  $1.204 \times 10^{-15}$  and the largest dynamics residual is  $6.492 \times 10^{-15}$ .

### 9.5. Coordinate-level mechanism specification

The setup table gives the physical scales; the following coordinate data make the lower-pair graphs reproducible directly from the manuscript. Let  $e_x = (1, 0, 0)$ ,  $e_y = (0, 1, 0)$ , and  $e_z = (0, 0, 1)$ . All point vectors  $s$  and direction vectors  $a$  are body-fixed unless the body is ground; each CD group uses the three coordinate directions  $e_x, e_y, e_z$ . Driver functions listed here are the functions actually used by the setup code after model-file overrides.

Table 17: Four-example validation summary.

| Example         | Accepted evidence                                  | Role                    | Headline metric                                                                                         |
|-----------------|----------------------------------------------------|-------------------------|---------------------------------------------------------------------------------------------------------|
| Single pendulum | Driven absolute-coordinate FullVA residual         | Dynamic-order evidence  | Minimum observed order 6.024; max stage residual $3.664 \times 10^{-12}$ .                              |
| Double pendulum | Method-side FullVA reference policy                | Dynamic-order evidence  | Minimum orientation/velocity order 6.089; endpoint velocity constraints below $8.323 \times 10^{-12}$ . |
| Four link       | Closed-loop kinematic FullVA and reaction dynamics | Mechanism coverage only | Max constraint norm $1.052 \times 10^{-14}$ ; max dynamics residual $1.338 \times 10^{-13}$ .           |
| Slider crank    | Closed-loop kinematic FullVA and reaction dynamics | Mechanism coverage only | Max constraint norm $1.204 \times 10^{-15}$ ; max dynamics residual $6.492 \times 10^{-15}$ .           |

Table 18: Problem setup values used by the four ASME-style validation examples.

| Example         | Physical and constraint setup                                                                                                                                                                                                                                        | Time grid and reference policy                                                                                                                                                        | Accepted check                                                                                                    |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Single pendulum | One steel bar, length 4 m, square side 0.05 m, density 7800 kg/m <sup>3</sup> , mass 78 kg, and $J = \text{diag}(0.0325, 104.016, 104.016)$ kg m <sup>2</sup> . The graph has one ground revolute joint: CD(3) and DP1(3).                                           | Driven angle $\theta(t) = \pi/2 + (\pi/4) \cos(2t)$ , analytic reference, $t_f = 0.2$ , and $h = \{0.20, 0.10, 0.05\}$ .                                                              | Minimum observed order 6.024; max stage residual $3.664 \times 10^{-12}$ .                                        |
| Double pendulum | Two steel bars, lengths 4 m and 2 m, same side and density as the single pendulum, masses 78 kg and 39 kg, with two revolute joints and no friction or external torque.                                                                                              | $q(0) = (-\pi/2, 0)$ and regularized angular speeds $(10^{-12}, -10^{-12})$ rad/s. The accepted reference is nested local FullVA at $h = 0.005$ ; tested $h = \{0.04, 0.02, 0.01\}$ . | Orientation and angular-velocity orders 6.101/6.089; endpoint velocity constraint below $8.323 \times 10^{-12}$ . |
| Four-link       | Three moving bodies with masses 2, 1, 1 kg and inertia diagonals $(4, 2, 0)$ , $(12.4, 0.01, 0)$ , and $(4.54, 0.01, 0)$ kg m <sup>2</sup> . The closed loop has CD(12) and DP1(6).                                                                                  | Driven constraint $\cos(\pi t + \pi/2)$ , $t_f = 0.2$ , $h = \{0.02, 0.01, 0.005\}$ , and local kinematic FullVA reference $h = 0.001$ .                                              | Max constraint norm $1.052 \times 10^{-14}$ ; max dynamics residual $1.338 \times 10^{-13}$ .                     |
| Slider-crank    | Crank, rod, and slider masses 0.12, 0.5, 2 kg with inertia diagonals $(10^{-4}, 10^{-5}, 10^{-4})$ , $(4 \times 10^{-3}, 4 \times 10^{-4}, 4 \times 10^{-3})$ , and $(10^{-4}, 10^{-4}, 10^{-4})$ kg m <sup>2</sup> . The graph has CD(6), DP1(7), DP2(4), and D(1). | Driven constraint $\cos(-2\pi t + \pi/2)$ , $t_f = 0.2$ , $h = \{0.02, 0.01, 0.005\}$ , and local kinematic FullVA reference $h = 0.001$ .                                            | Max constraint norm $1.204 \times 10^{-15}$ ; max dynamics residual $6.492 \times 10^{-15}$ .                     |

*Single pendulum.* The moving body is body 1 and ground is body 0. The ground revolute joint uses CD rows with  $s_1 = (-2, 0, 0)$  and  $s_0 = (0, 0, 0)$ , plus DP1 rows  $(a_1, a_0) = (e_x, e_x)$  and  $(e_y, e_x)$ . The driven DP1 row uses  $(a_1, a_0) = (e_y, -e_z)$  and  $f(t) = \cos(\pi/2 + (\pi/4)\cos(2t))$ .

*Double pendulum.* The two links are bodies 1 and 2, with ground body 0. The ground revolute joint for body 1 uses CD rows with  $s_1 = (-2, 0, 0)$  and  $s_0 = (0, 0, 0)$ , plus DP1 rows  $(e_x, e_x)$  and  $(e_y, e_x)$ . The inter-link revolute joint uses CD rows between body 1 and body 2 with  $s_1 = (2, 0, 0)$  and  $s_2 = (-1, 0, 0)$ . The second axis pair is represented by DP1 rows on body 2 against ground:  $(a_2, a_0) = (e_x, e_x)$  and  $(e_y, e_x)$ .

*Four-link closed loop.* The moving bodies are the rotor, link 2, and link 3, numbered 1, 2, and 3; ground is body 0. Revolute joint A between body 1 and ground uses CD rows with  $s_1 = (0, 0, 0)$  and  $s_0 = (0, 0, 0)$ , plus DP1 rows  $(e_x, e_x)$  and  $(e_y, e_x)$ . Revolute joint D between body 3 and ground uses CD rows with  $s_3 = (0, 3.7, 0)$  and  $s_0 = (-4, -8.5, 0)$ , plus DP1 rows  $((0, 0, 1), e_y)$  and  $(e_y, e_y)$ . The spherical joint between bodies 3 and 2 uses CD rows with  $s_3 = (0, -3.7, 0)$  and  $s_2 = (0, -6.1, 0)$ . The universal joint between bodies 1 and 2 uses CD rows with  $s_1 = (-2, 0, 0)$  and  $s_2 = (0, 6.1, 0)$ , plus the DP1 row  $((0, -0.662, 0.75), e_z)$ . The driving DP1 row uses  $((-1, 0, 0), e_y)$  and  $f(t) = \cos(\pi t + \pi/2)$ .

*Slider-crank closed loop.* The crank, connecting rod, and slider are bodies 1, 2, and 3; ground is body 4. Revolute joint A between ground and crank uses CD rows with  $s_4 = (0, 0.1, 0.12)$  and  $s_1 = (0, 0, 0)$ , plus DP1 rows  $(e_y, e_x)$  and  $(e_z, e_x)$ . The spherical joint B between rod and crank uses CD rows with  $s_2 = (-0.15, 0, 0)$  and  $s_1 = (0, 0.08, 0)$ . The revolute/cylindrical joint C between rod and slider uses one DP1 row  $(e_y, e_z)$  and two DP2 rows with rod point  $s_2 = (0.15, 0, 0)$ , slider point  $s_3 = (0, 0, 0)$ , and rod directions  $e_y$  and  $e_z$ . The translational joint D between slider and ground uses DP1 rows  $(-e_x, e_x)$ ,  $(e_y, e_x)$ , and  $(-e_x, e_y)$ , plus DP2 rows from slider point  $s_3 = (0, 0, 0)$  to ground point  $s_4 = (0.2, 0, 0)$  with slider directions  $-e_x$  and  $e_y$ . The distance row constrains the rod point  $(0.15, 0, 0)$  and slider point  $(0, 0, 1)$  with  $f(t) = 1$ . The driving DP1 row uses  $(e_y, e_y)$  and  $f(t) = \cos(-2\pi t + \pi/2)$ .

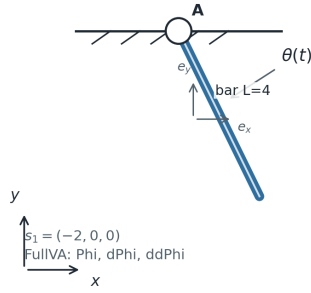
## 10. Lower-pair constraint formulas

This section makes explicit the scalar constraint rows used by the four example mechanisms. Let body  $i$  have center position  $r_i$ , rotation matrix

## ASME-style lower-pair validation mechanisms

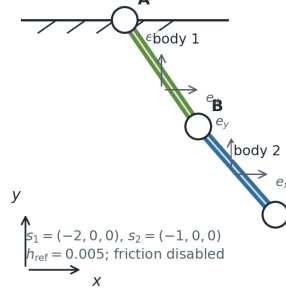
### single pendulum

CD(3) + DP1(3); driven theta(t)



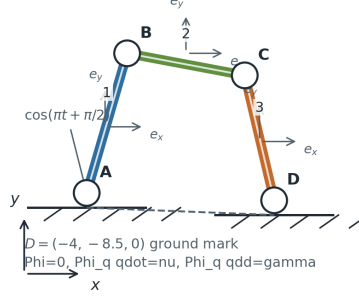
### double pendulum

CD(6) + DP1(4); nested FullVA reference



### four-link loop

CD(12) + DP1(6); driven closed loop



### slider-crank

CD(6) + DP1(7) + DP2(4) + D(1)

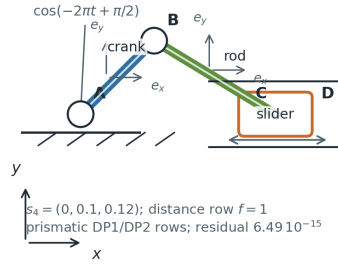


Figure 3: Coordinate-oriented mechanism schematics and lower-pair constraint inventory for the four ASME-style examples. Each panel shows the global coordinate frame, named joints, the principal moving bodies, the driver where applicable, and the CD/DP1/DP2/D constraint vocabulary used by the FullVA rows. Exact body-fixed point and direction vectors are listed in the text immediately preceding the figure.

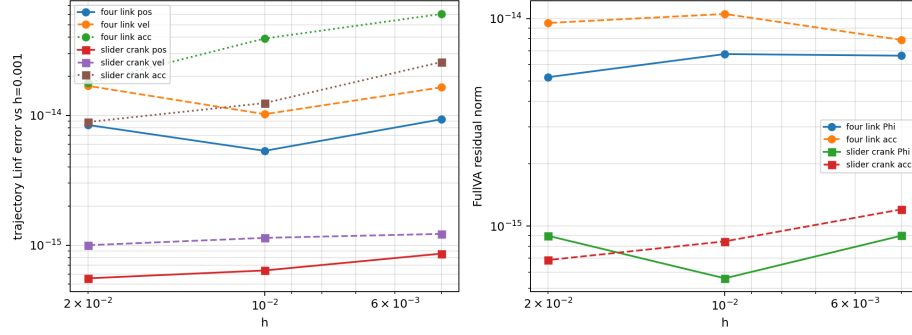


Figure 4: Closed-loop kinematic FullVA mechanism-coverage evidence for the four-link and slider-crank examples.

$A_i \in SO(3)$ , translational velocity  $v_i$ , angular velocity  $\omega_i$ , translational acceleration  $a_i$ , and angular acceleration  $\alpha_i$ . A marked point and a marked direction fixed in body  $i$  are  $x_i = r_i + A_i s_i$  and  $u_i = A_i a_i$ , respectively. The row families used in the examples are listed in Eq. (48):

$$C_{CD,k}(q, t) = e_k^T (x_j - x_i) - d_k(t), \quad (45)$$

$$C_{DP1}(q, t) = u_i^T u_j - c(t), \quad (46)$$

$$C_{DP2}(q, t) = u_i^T (x_j - x_i) - d(t), \quad (47)$$

$$C_D(q, t) = \|x_j - x_i\|^2 - \ell^2(t). \quad (48)$$

Here CD denotes coordinate difference, DP1 constrains two body-fixed directions, DP2 constrains a body-fixed direction against a point difference, and D is a distance row. Revolute joints are assembled from CD and DP1 rows; the slider-crank prismatic joint additionally uses DP2 and distance rows.

The velocity and acceleration formulas use the rigid-body identities in Eq. (50):

$$\dot{x}_i = v_i + \omega_i \times A_i s_i, \quad \dot{u}_i = \omega_i \times u_i, \quad (49)$$

$$\ddot{x}_i = a_i + \alpha_i \times A_i s_i + \omega_i \times (\omega_i \times A_i s_i), \quad \ddot{u}_i = \alpha_i \times u_i + \omega_i \times (\omega_i \times u_i). \quad (50)$$

Writing  $\Delta x = x_j - x_i$ ,  $\Delta \dot{x} = \dot{x}_j - \dot{x}_i$ , and  $\Delta \ddot{x} = \ddot{x}_j - \ddot{x}_i$ , the row derivatives

assembled by the FullVA block are Eq. (54) at velocity level

$$\dot{C}_{\text{CD},k} = e_k^T \Delta \dot{x} - \dot{d}_k, \quad (51)$$

$$\dot{C}_{\text{DP1}} = \dot{u}_i^T u_j + u_i^T \dot{u}_j - \dot{c}, \quad (52)$$

$$\dot{C}_{\text{DP2}} = \dot{u}_i^T \Delta x + u_i^T \Delta \dot{x} - \dot{d}, \quad (53)$$

$$\dot{C}_{\text{D}} = 2\Delta x^T \Delta \dot{x} - 2\ell \dot{\ell}, \quad (54)$$

and Eq. (58) at acceleration level:

$$\ddot{C}_{\text{CD},k} = e_k^T \Delta \ddot{x} - \ddot{d}_k, \quad (55)$$

$$\ddot{C}_{\text{DP1}} = \ddot{u}_i^T u_j + 2\dot{u}_i^T \dot{u}_j + u_i^T \ddot{u}_j - \ddot{c}, \quad (56)$$

$$\ddot{C}_{\text{DP2}} = \ddot{u}_i^T \Delta x + 2\dot{u}_i^T \Delta \dot{x} + u_i^T \Delta \ddot{x} - \ddot{d}, \quad (57)$$

$$\ddot{C}_{\text{D}} = 2\Delta \dot{x}^T \Delta \dot{x} + 2\Delta x^T \Delta \ddot{x} - 2(\dot{\ell}^2 + \ell \ddot{\ell}). \quad (58)$$

## 11. Diagnostic boundaries and limitations

This section states diagnostic boundaries; it is not a second contribution line and not a prerequisite for the theorem or the accepted validation rows. The method claim can be evaluated from the residual definition, Theorem 1, and Section 8; the diagnostics below are retained only to prevent stronger source-paper, full-TFE, sparse-performance, and external-superiority claims from being read into the main method evidence. The supplementary evidence contains additional diagnostic work toward source-paper temporal finite-element residual replacement. This evidence is useful because it identifies why a stronger reproduction claim is outside the present scope, but it is not needed for Proposition 1. The central diagnostic result is that several endpoint or weak-row variants can close one property at a time, but the reported evidence does not provide a source-free, projection-free, non-terminal-row-replacement 132-row TFE residual that both closes terminal endpoint velocity and preserves the smooth order target.

Sparse automatic differentiation is also treated as diagnostic engineering evidence rather than a primary paper claim. The accepted sparsity pattern is correct and quantified, but the recorded wall-clock measurements do not show a stable speed win over dense `jacfwd`. The current code should therefore be read as a reference implementation. The recorded small-system pattern has a  $132 \times 132$  dense dimension, 2637 nonzeros, and 90/60 column/row

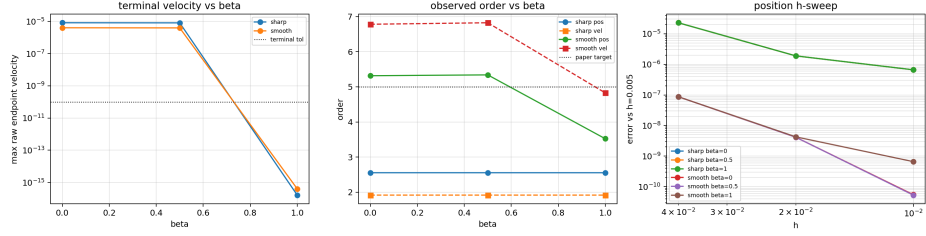


Figure 5: Order/closure blend diagnostic. The tested path separates order preservation from terminal endpoint-velocity closure, which is why independent full-TFE replacement remains open.

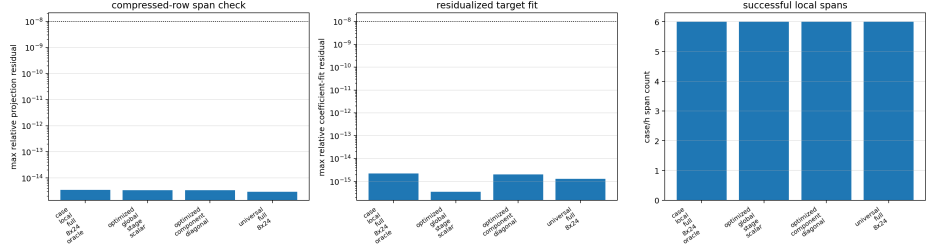


Figure 6: Velocity-compression diagnostic. Local row-space probes quantify the gap between available weak-row features and the terminal bridge needed for accepted full-TFE stage replacement.

colors; row-VJP is still 1.20–1.22 times the dense runtime, requiring a 16.9–18.3% row-runtime reduction merely to match dense. No body-count sweep for  $N = 2, 4, 8, 16, 32$ , memory scaling, conditioning-versus-body-count, or Jacobian/linear-solve breakdown is claimed.

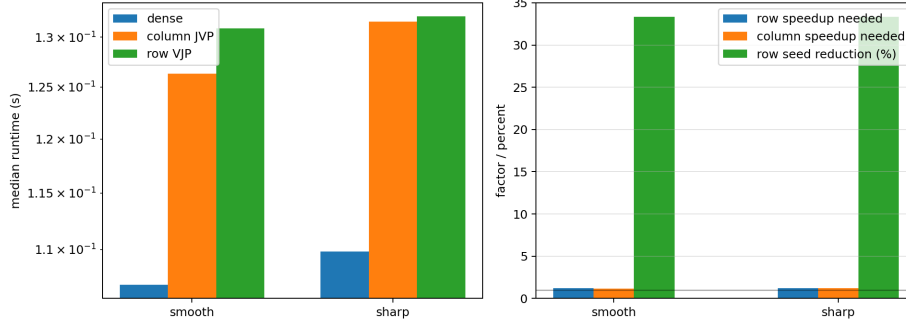


Figure 7: Sparse backend speed-gap diagnostic. The sparse structure is correct and quantified, but dense `jacfwd` remains faster in the recorded wall-clock evidence.

## 12. Limitations

Six limitations define the boundary of the claim, ordered from theorem and method conditions to external reproduction boundaries.

1. **Conditional proof boundary.** The stage-residual input of the theorem is closed exactly: by Lemma 1 the lifted reduced Gauss stage satisfies all 132 implemented rows, so no stage-residual perturbation term appears. The theorem remains conditional on the smooth regularity tube, the local inverses and stability scale of P2 (derived from P1 for small steps in Lemma 3), and an inexact nonlinear solve satisfying the stated compact-tube  $\eta_h^{\text{tube}} \leq c_\eta h^7$  policy; residual-to-error promotion for additional four-link/slider-crank dynamic-order claims remains outside the accepted theorem. Promoting closed-loop residual or reaction rows to trajectory order would require a separate branch-compatible residual-to-error theorem with a fixed residual operator, trajectory norm, stability or inf-sup inverse, nonlinear remainder control, reference-floor exclusion, and its own dynamic-order campaign. Small constraint, reaction, or mechanism residuals alone do not provide that theorem; P7 remains a separate output nonclaim/residual-to-error boundary.

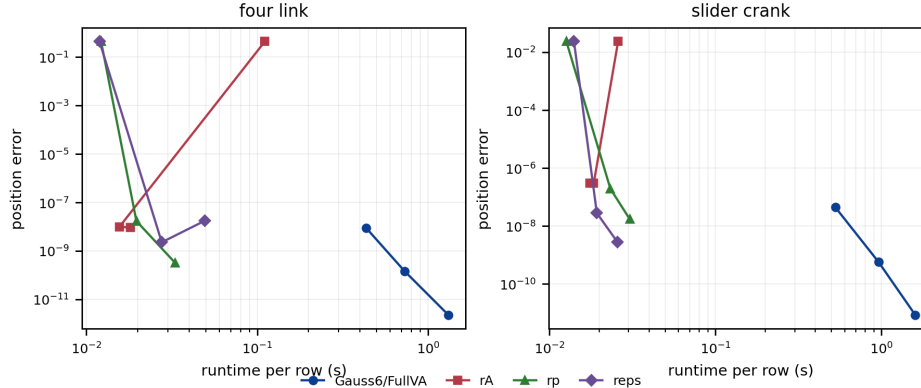


Figure 8: Strict common-reference work/precision rows for bounded diagnostics on the closed-loop four-link and slider-crank mechanisms. Bounded common-reference diagnostic work/precision rows use the same finite accepted-method endpoint reference at  $h = 0.0125$  over the coarse  $h = 0.1, 0.05, 0.025$  window. The plot removes one fixed-grid diagnostic caveat for these coarse rows, but it is still not an external superiority claim because the broader source-paper suites and manuscript-level review remain open.

Table 19: Diagnostic boundaries separated from the accepted theorem.

| Diagnostic boundary              | Current evidence                                                                                                        | Claim boundary     |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------|--------------------|
| Independent full-TFE replacement | No accepted 132-row paper-derived weak residual with source-free terminal closure and smooth order recovery.            | Boundary retained. |
| Sparse backend speed             | Sparse pattern and colors are quantified, but dense <code>jacfwd</code> is still faster in the recorded wall-clock run. | Backend caveat.    |
| Sharp-friction coarse regime     | Coarse sharp-friction sweep is order-reduced; deeper fixed refinement recovers high order at increased cost.            | Regime caveat.     |

| Claim boundary and remaining limitations                                                                                                                                 |                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| accepted method claim                                                                                                                                                    | four-example evidence split                                                                                                                  |
| <b>accepted</b> <b>Gauss6/FullVA order</b><br>conditional sixth-order path; observed smooth slopes 7.161/7.066 are finite-window support, not a seventh-order theorem    | <b>✓ single</b><br>dynamic order accepted                                                                                                    |
| <b>accepted</b> <b>mechanism coverage</b><br>four ASME-style mechanisms are covered; dynamic order is accepted only for single and double pendula                        | <b>✓ double</b><br>dynamic order accepted                                                                                                    |
| <b>bounded</b> <b>source-paper comparison</b><br>local $m=3$ TFE formula target has expected order five; complete source-paper residual replacement is outside the claim | <b>! four-link</b><br>coarse dynamics / common-ref diagnostics; residual-to-error theorem open                                               |
| <b>open</b> <b>external superiority</b><br>not claimed until the same-test public-code campaign and work metrics are complete                                            | <b>! slider-crank</b><br>coarse dynamics / common-ref diagnostics; residual-to-error theorem open                                            |
| proof and solver limitations                                                                                                                                             | same-test benchmark limitations                                                                                                              |
| <b>conditional</b> <b>conditional theorem</b><br>requires regular reduced flow, smooth lift, isolated Newton solve, and stage residual defect $O(h^7)$                   | <b>open</b> <b>original TFE pendulum</b><br>source-paper pendulum rows remain to be reproduced on the same policies                          |
| <b>direct route</b> <b>implementation oracle</b><br>runtime layout, AD-expanded oracle, and direct dynamic row closure pass; primitive symbolic lane remains diagnostic  | <b>partial</b> <b>2021 rA/rp/reps suite</b><br>public baselines and coarse pilots exist; full dynamic work/precision comparison remains open |
| <b>open</b> <b>nonlinear tolerance</b><br>proof-level policy is $\eta_h = O(h^7)$ ; fixed absolute tolerances are finite-run evidence                                    | <b>partial</b> <b>2022 half-implicit suite</b><br>bounded pilot is present; full $T=8$ policy and method rows remain open                    |
| <b>open</b> <b>residual-to-error route</b><br>closed-loop residual rows are not promoted to order claims without the missing theorem                                     | <b>open</b> <b>velocity partitioning</b><br>code path unresolved in the local public-code mirror and metadata search                         |

Figure 9: Claim-boundary and limitation map for the present paper. Green items are accepted claims or accepted evidence rows; orange items are bounded or partial evidence; red items remain open boundaries. The figure summarizes why the manuscript supports a conditional sixth-order Gauss6/FullVA claim under the retained theorem interfaces but still does not claim complete source-paper residual replacement, external same-test superiority, or a closed residual-to-error transfer theorem for the closed-loop mechanisms.

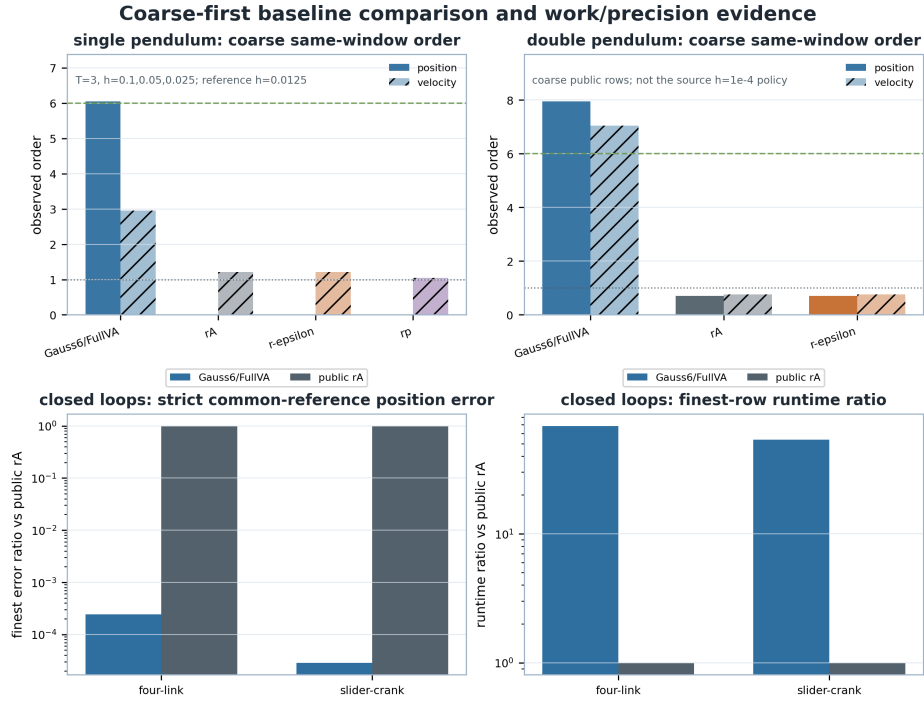


Figure 10: Coarse-first baseline and work/precision diagnostics. The top panels compare observed order on the same coarse window for the single and double pendula. The lower panels use the strict common-reference closed-loop summaries for four-link and slider-crank, with all plotted rows evaluated against the same finite accepted-method endpoint reference at  $h_{\text{ref}} = 0.0125$ . These panels document coarse-first diagnostics and work/precision availability, but they are not an external-superiority claim and do not replace the unresolved original source-policy campaigns.

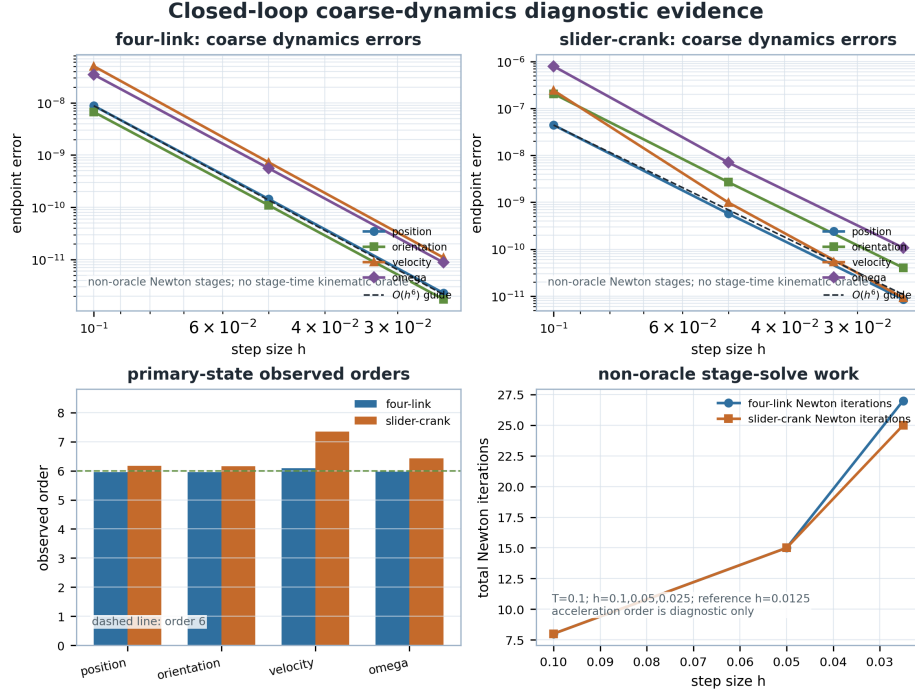


Figure 11: Closed-loop coarse-dynamics diagnostic evidence for the four-link and slider-crank mechanisms. Closed-loop coarse-window trajectory diagnostics plot endpoint position, orientation, linear-velocity, and angular-velocity errors against the  $O(h^6)$  guide over  $h = 0.1, 0.05, 0.025$  with  $h_{\text{ref}} = 0.0125$ . The bottom panels summarize the finite-window fitted slopes and the non-oracle Newton stage-solve work. These rows are mechanism-coverage and coarse-window trajectory diagnostics only, not accepted dynamic-order proof; they run without using stage-time kinematic oracles, without making  $10^{-4}$  a default policy, without promoting four-link or slider-crank to accepted dynamic-order rows, and without asserting external same-test superiority. They are finite-window diagnostics: they do not instantiate the residual-to-error implication, do not close P7, and do not promote four-link or slider-crank to accepted dynamic-order examples.

### All-method common-reference result matrix

Each cell reports observed velocity order  $p$  and finest-step velocity error  $e$ .

|               |            | Single                    | Double                     | Four-link                  | Slider-crank               |
|---------------|------------|---------------------------|----------------------------|----------------------------|----------------------------|
| Gauss6/FullVA | internal   | $p=5.80$<br>$e=1.2e-12$   | $p=6.01$<br>$e=2.3e-13$    | $p=6.08$<br>$e=1.1e-11$    | $p=7.34$<br>$e=9.2e-12$    |
| HI22 rA       | diagnostic | $p=1.00^*$<br>$e=7.7e-02$ | $p=1.05^*$<br>$e=4.9e-05$  | $p=1.02^*$<br>$e=2.0e-01$  | $p=0.94^*$<br>$e=3.3e-02$  |
| HI22 half     | diagnostic | $p=1.84^*$<br>$e=7.7e-02$ | $p=-0.80^*$<br>$e=3.0e+00$ | $p=1.59^*$<br>$e=2.0e-01$  | $p=2.46^*$<br>$e=3.3e-02$  |
| RA21 rA       | diagnostic | $p=0.62^*$<br>$e=1.3e+01$ | $p=1.05^*$<br>$e=4.9e-05$  | $p=1.02^*$<br>$e=2.0e-01$  | $p=0.94^*$<br>$e=3.3e-02$  |
| RA21 reps     | diagnostic | $p=0.62^*$<br>$e=1.3e+01$ | $p=1.05^*$<br>$e=4.9e-05$  | $p=2.89^*$<br>$e=1.5e-02$  | $p=0.94^*$<br>$e=3.3e-02$  |
| RA21 rp       | diagnostic | $p=0.52^*$<br>$e=1.5e+01$ | $p=0.36^*$<br>$e=5.2e-05$  | $p=2.89^*$<br>$e=1.5e-02$  | $p=0.94^*$<br>$e=3.3e-02$  |
| TFE Newmark   | diagnostic | $p=0.90^*$<br>$e=1.5e+01$ | $p=1.79^*$<br>$e=4.2e-04$  | $p=1.87^*$<br>$e=1.7e-02$  | $p=2.05^*$<br>$e=1.5e-03$  |
| TFE m1        | diagnostic | $p=1.02^*$<br>$e=1.5e+01$ | $p=2.00^*$<br>$e=2.7e-04$  | $p=3.87^*$<br>$e=5.3e-04$  | $p=2.48^*$<br>$e=6.3e-04$  |
| TFE m2        | diagnostic | $p=0.05^*$<br>$e=1.1e+02$ | $p=1.87^*$<br>$e=8.7e-05$  | $p=1.02^*$<br>$e=6.7e-04$  | $p=2.29^*$<br>$e=9.9e-05$  |
| TFE trap.     | diagnostic | $p=1.02^*$<br>$e=1.5e+01$ | $p=2.00^*$<br>$e=2.7e-04$  | $p=3.89^*$<br>$e=5.3e-04$  | $p=2.33^*$<br>$e=7.3e-04$  |
| VP24 rA proxy | diagnostic | $p=0.00^*$<br>$e=3.2e-11$ | $p=1.00^*$<br>$e=2.2e-03$  | $p=-0.00^*$<br>$e=1.1e-08$ | $p=-0.00^*$<br>$e=1.3e-07$ |

p >= 5.5
3 <= p < 5.5
1.5 <= p < 3
0.5 <= p < 1.5
p < 0.5

\* marks a row with diagnostic/source-policy caveats. Nonlocal rows are common-reference diagnostics only; they are not source-paper superiority claims.  
Source: PAPER\_NUMERICAL\_RESULT\_MATRIX.json;  $h = 0.1, 0.05, 0.025$ ; reference  $h = 0.0125$ ; 44/44 method-example cells.

Figure 12: All-method common-reference result matrix for the four validation examples. Each cell reports observed velocity order and finest-step velocity error from the paper numerical result matrix over  $h = 0.1, 0.05, 0.025$  with  $h_{\text{ref}} = 0.0125$ . The local Gauss6/FullVA row is method-side order evidence. All nonlocal rows remain bounded diagnostic rows with open source-policy status, so the figure does not authorize an external source-paper superiority claim.

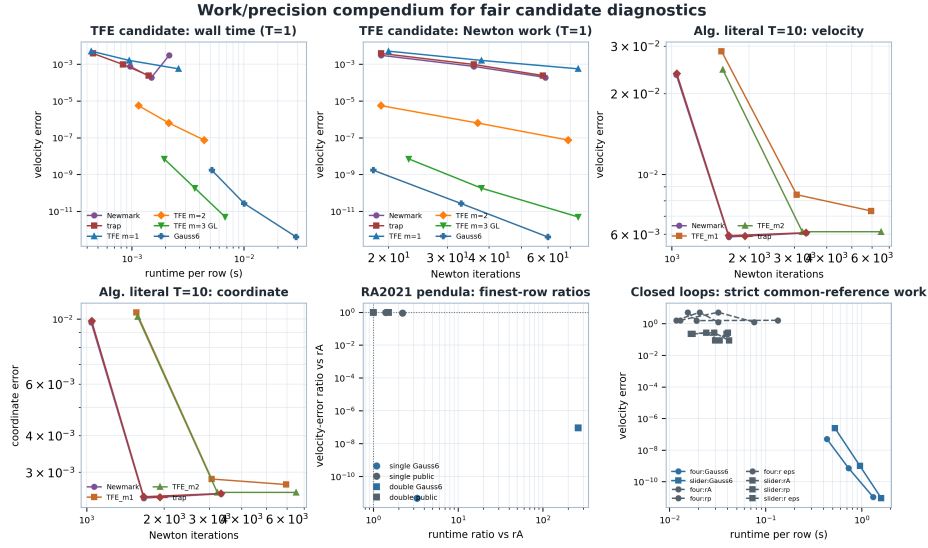


Figure 13: Work/precision compendium for fair candidate diagnostics. The first two panels use the frictionless TFE source-pendulum candidate route on a shared  $T = 1$  grid and RK4 reference, plotting velocity error against wall time and Newton work. The Algorithm-1-literal  $T = 10$  panels plot velocity and coordinate error against Newton work from the Algorithm-1-literal work/precision diagnostic rows; these fixed- $h$  endpoint rows remain source-policy open because exact- $T$  error sampling and a source-equivalent DAE/method runner are not established. The RA2021 panel summarizes same-window pendulum finest-row runtime and velocity-error ratios relative to public  $rA$  rows, and the closed-loop panel collects strict common-reference four-link and slider-crank work/precision rows. These panels consolidate existing evidence for review; they remain candidate/common-reference diagnostics, not source-policy superiority claims.

2. **Nonlinear tolerance and endpoint closure.** The asymptotic theorem requires a compact-tube nonlinear residual policy  $\eta_h^{\text{tube}} \leq c_\eta h^7$ , while the reported production runs use a fixed finite-run tolerance. The current scaled-tolerance and tolerance-regime probes are finite reported-window diagnostics around this policy; they diagnose this scale but do not close it, prove an asymptotic solver contract, or convert fixed production tolerances into theorem hypotheses. Likewise, endpoint closure is a conditional local  $O(h^7)$  perturbation under Lemma 5 and the retained P2 endpoint-ball/right-inverse hypotheses; the raw residual/correction rows do not prove those retained hypotheses and are not hidden source-paper or reference-solution projections. Lemma 7 supplies the solver result that closing P6 demands on the same compact tube: under P1 and P2 and for  $h \leq h_0$  the lifted endpoint predictor lies in the strong local residual-inverse ball, the Newton iterates from any point of that ball stay in it, and the  $h^7$ -scaled stopping rule is met after  $O(\log(1/h))$  iterations with constants depending only on the tube. P6 is nevertheless retained as a hypothesis: the Algorithm 1 predictor is not the lifted endpoint predictor and its ball membership is confirmed only a posteriori (Table 12), the reported production runs use a fixed tolerance rather than that rule, so they are covered only on transitions where the accepted residual meets the envelope, and the Neumann threshold  $h_0$  is far below the reported step sizes in the implemented norm. Discharging P6 would in any case address the solver-scale interface only; it would still not promote residual-only mechanism rows unless the separate P7 residual-to-error transfer theorem were also proved.
3. **Full source-paper residual replacement.** Independent full-TFE stage replacement is not accepted. The accepted theorem is tied to the Gauss/FullVA residual in Eq. (6); replacing those rows by paper-derived weak rows is a separate method-construction problem.
4. **Performance and scalability.** The sparse automatic differentiation structure is correct and quantified, but dense `jacfwd` remains faster in the recorded wall-clock evidence. The sparse backend is therefore an implementation diagnostic, not a claimed performance improvement. The current implementation is a small-system reference implementation; a production scalability claim would require an  $N = 2, 4, 8, 16, 32$  chain benchmark with wall-clock time, Newton iterations, Jacobian-assembly time, linear-solve time, conditioning or rank diagnostics, and

memory scaling.

5. **Sharp-friction coarse regime.** The practical coarse sharp-friction regime is order-reduced, although deeper fixed refinement recovers high order at additional cost. The method claim therefore rests on the smooth contract and treats sharp friction as a quantified caveat.
6. **External same-test comparison.** The exact same-test external comparison criterion in Section Appendix A remains open after the public-code baselines and selected/coarse Gauss6/FullVA pilots. The Chaturvedi–Sandu–Sandu pendulum tests are not source-policy reproduced: the encoded source-shaped candidate smokes and bounded probes remain diagnostic; the Kissel/Negrut suites still lack complete Gauss6/FullVA dynamic order/work rows under the published policies; and the closed-loop surrogate rows do not change that accepted-order boundary. The floor analysis narrows this boundary to velocity/acceleration floor evidence plus a position/reference floor, but it does not close the source-policy dynamic-order comparison. The velocity-partitioning public-code path is also unresolved.

These limitations do not invalidate the narrower conditional formal-order comparison; they define the proof, implementation, and reproduction work still needed before a broader paper-level claim would be justified.

### 13. Conclusions

This paper has presented a monolithic Lie-group Gauss6/FullVA integrator for lower-pair mechanisms. The method combines  $SO(3)$  retractions, lower-pair position/velocity/acceleration rows, and Newton–Euler dynamics in one square stage residual, then reconstructs the endpoint on the accepted branch. Under the stated compact-branch, endpoint/stability, and solver-scale contracts, this construction gives the conditional sixth-order Lie-group FullVA theorem of Theorem 1. The conditional qualifier fixes the compact branch, endpoint, and solver-scale domain of the theorem; it is not an empirical replacement for the local-defect proof. The numerical evidence records finite-window smooth position and velocity orders 7.161 and 7.066 as consistency evidence after the order-six theorem has been stated. The single- and double-pendulum rows provide method-side dynamic-order evidence, while the four-link and slider-crank rows provide diagnostic mechanism-coverage evidence, not accepted dynamic-order proof and not residual-to-error evidence, for the same FullVA residual. Against the local order-five  $m = 3$

Gauss–Lobatto temporal finite-element target, this is a conditional formal-order comparison, not an external same-test superiority claim.

The proof route is anchored by the exact stage identity: the lifted reduced Gauss stage satisfies all 132 implemented rows, so the accepted stage solve is three-stage Gauss collocation of the reduced joint-coordinate equation of motion written in absolute coordinates. The theorem is then a conditional implication for a fixed smooth branch: once the compact branch, local inverses, endpoint map, stability scale, and  $\eta_h^{\text{tube}} \leq c_\eta h^7$  solver-scale condition are in force, the Gauss endpoint defect, the endpoint-closure perturbation, and the inexact-Newton perturbation supply the  $O(h^7)$  local defect, and the Gronwall transfer gives the  $O(h^6)$  same-initial-state reported-grid estimate. The finite-window order fits, finite solver probes, and residual/reaction tables are consistency or diagnostic evidence and do not replace those mathematical inputs. Remaining work concerns a separate uniform same-branch solver-policy theorem, residual-to-error promotion for closed-loop residual rows, and external same-test reproduction; these are outside the theorem proved here.

Broader comparison work remains separate from this method result. It should rerun the original pendulum tests and the Kissel/Negrut public-code mechanism suites under identical benchmark definitions, reporting order, error, constraint drift, Newton work, and runtime side by side. The velocity-partitioning code path is part of that separate published-test baseline scope. Complete source-paper temporal finite-element residual replacement and a runner-centered full reproduction package remain outside the accepted claim.

### **CRedit authorship contribution statement**

Jingquan Wang: Conceptualization, Methodology, Software, Validation, Formal analysis, Writing - original draft, Writing - review and editing.

### **Declaration of competing interest**

The author is affiliated with the University of Wisconsin–Madison. The author declares no competing interests relevant to this work.

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## Data availability

The journal source archive is the LaTeX archive: it contains the manuscript sources, figures, and declarations. This is an editorial-source archive rather than a full source-policy runner archive. The companion reproducibility package contains the reported-result records and read-only check scripts used for this paper, together with the Lean 4 development that machine-checks the lemmas of Section 7. It is scoped to the accepted method-side evidence and proof-dependency records; external same-test runner reproduction for source papers and full TFE runner comparisons are reserved for a separate package. An archival public repository or DOI can be attached at journal production or revision if the editorial office requires a public deposit.

## Declaration of generative AI and AI-assisted technologies in the writing process

During preparation of this work, the author used an AI coding assistant for drafting support, consistency checking, and validation-script editing. The author reviewed and edited the content and takes full responsibility for the published article.

## Appendix A. Cross-paper same-test comparison criterion: diagnostic appendix

This subsection is a diagnostic guardrail, not a second contribution claim and not part of the proof chain. It can be read as a promotion rule for separate external source-paper claims after the method definition, proof, and accepted method-side validation have already been fixed. A publication-level source-paper comparison must be made on the same numerical tests used by the source papers, not only on mechanisms with similar names. The criterion is strict: the body parameters, initial conditions, prescribed drivers, friction model, step-size sweep, nonlinear tolerance policy, reference solution, error norm, and work metric must be matched or explicitly justified. Table A.20 records the required external comparisons, the status of this evidence package, and the claim that is allowed before the remaining source-suite decisions close. The current paper uses these rows only to prevent overclaiming: the method claim, the conditional theorem, and the accepted numerical evidence do not depend on source-policy promotion.

*Why the source-policy criterion is separate.* The source-policy criterion is required only for a stronger external apples-to-apples statement: that Gauss6/FullVA has been evaluated under the source papers' own model, output, tolerance, reference, and work policies and would then be eligible for a separate source-paper replacement or superiority comparison. It is not a premise of the Gauss6/FullVA residual definition, the retained conditional sixth-order theorem, or the accepted local dynamic-order and mechanism-coverage evidence. Therefore an open source-policy row does not weaken the bounded method/proof claim; it only blocks the broader external same-test and full reproducibility package claims.

*Source-policy progress boundary.* The detailed source-policy record is boundary evidence only. It records local candidate availability, public-code setup progress, TFE formula diagnostics, and open runner gaps, but no external row is promoted to a source-policy dynamic-order or source-paper superiority claim. The current progress entries are read only as diagnostic inventory: public-code baseline order/timing rows, source-identity checks, coarse double-pendulum self-reference runs, bounded four-link/slider-crank mechanism-diagnostic candidates, and published-formula candidates record implementation progress but do not establish external same-test reproduction. No external source-policy dynamic-order row is accepted, and the local method rows under external source-policy dynamic-order rules remain unpromoted; therefore no source-paper superiority, full source-policy work-precision, or full-TFE replacement claim is made.

*Claim-policy implication.* The source-policy promotion criterion is interpreted as a logical promotion rule, not as a visual preference over plots. A paper-level external-superiority statement requires the source benchmark, method-set, numerical-control, and work-metric rows in Table A.21 to close for the source-policy campaign being cited. The current evidence record reports no strict external error-claim rows, does not establish any source-policy-closed external same-test error row, and leaves the same-test promotion conditions open. Therefore the common-reference matrix and bounded pilots may be used only as diagnostics and cannot be used as premises in any implemented source-paper superiority claim.

Consequently, the present numerical evidence supports only the method-side Gauss/FullVA order claim and the local order-five TFE formula-target comparison. The method-side evidence consists of the smooth cylindrical-chain benchmark plus the four ASME-style mechanisms in this package; the

Table A.20: Required same-test comparisons and current external-suite claim boundary.

| External source                                                              | Published or code-backed tests                                                                                              | Current evidence                                                                                                                                                                                                                                                                                          | Claim allowed now                                                               |
|------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Chaturvedi–Sandu–Sandu Lie-group TFE paper [1]                               | Rigid pendulum without and with joint friction; $m = 1, 2, 3$ TFE; Newmark– $\beta$ and trapezoidal comparators.            | Specification and source-shaped candidate smokes are encoded, including source parameter/output policies and bounded TFE probes; source-policy absolute-coordinate DAE/method-runner rows are outside the current evidence scope.                                                                         | Formula-order comparator only; no implemented source-paper superiority claim.   |
| Kissel–Taves–Negrut absolute-coordinate $rA/rp/r\epsilon$ suite [11, 12, 13] | Single pendulum, double pendulum, slider-crank, and four-link mechanisms from the public C1/C2 Python code and model files. | Public baseline rows and timing rows are complete. Gauss6/FullVA has single-pendulum public rows, double-pendulum coarse rows, and closed-loop strict-common-reference work/precision rows, but accepted local source-policy dynamic-order evidence is not promoted for any of the four tracked examples. | Coarse-first and common-reference evidence only; no external superiority claim. |
| Fang–Kissel–Zhang–Negrut half-implicit suite [14]                            | Double pendulum, slider-crank, four-link, $N$ -pendulum scaling, and slider-crank joint-friction examples.                  | Bounded $T = 0.1$ $rA/rA_{\text{half}}$ pilot complete; full $T = 8$ policy and Gauss6/FullVA rows open.                                                                                                                                                                                                  | Bounded pilot only; not a full half-implicit policy comparison.                 |
| Kissel–Bakke–Negrut velocity-partitioning Lie-group ODE suite [15]           | Single pendulum, double pendulum, four-link, and slider-crank tests reported for velocity partitioning.                     | The coordinate-partitioning proxy is identified through the implemented $rA$ wrapper; a distinct public VP code path remains unresolved and demoted. The source-code path is not separately reproduced.                                                                                                   | Common-reference diagnostic rows only; no source-policy VP superiority claim.   |

Table A.21: Minimum same-test TFE acceptance protocol before any source-paper superiority claim.

| Promotion item     | Required evidence                                                                                                                                                                                                               | Evidence boundary                                                                         |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Source benchmarks  | Reproduce the Chaturvedi–Sandu–Sandu frictionless pendulum and revolute joint with Brown–McPhee friction using the source parameters, horizon, initial data, drivers, and output variables.                                     | Specification and reference-feasibility diagnostics only; source rows not accepted.       |
| Method set         | Run source-equivalent TFE $m = 1, 2, 3$ , Newmark- $\beta$ , trapezoidal, and Gauss6/FullVA on the same trajectories.                                                                                                           | Local formula/common-reference diagnostics only; no accepted source-policy method matrix. |
| Numerical controls | Use the same step sizes or a justified common refinement, matched nonlinear tolerance policy, matched reference solution, and the same error norm and constraint-drift reporting.                                               | Open.                                                                                     |
| Work metrics       | Report error versus wall-clock time, Newton iterations, Jacobian assembly time, linear-solve time, and failed-step counts.                                                                                                      | Open.                                                                                     |
| Allowed claim      | Only after the rows above pass may the manuscript state a source-paper accuracy or work-precision advantage. Until then, the TFE comparison is a formal-order-level comparison only; diagnostic rows remain non-claim evidence. | No external-superiority claim.                                                            |

coverage matrix records 48 rows and 32 completed rows, but no external superiority claim. It does not support a stronger statement that Gauss6/FullVA is better than the published Chaturvedi–Sandu–Sandu implementation or the Kissel/Negrut public-code baselines on their own numerical tests. The extracted public-code benchmark specification and row-level inventory are maintained in the reproducibility appendix. The current evidence state is: specification extracted; 2021 public baselines complete for the already extracted order/timing evidence; 2021 public baseline order/timing evidence closed at 12/12 public order groups and 12/12 timing rows; selected/coarse Gauss6/FullVA pilot rows; a coarse public double-pendulum same-window baseline run; and a single-pendulum coarse same-window tranche. The accepted local Gauss6/FullVA source-policy dynamic-order set is nevertheless empty across the four tracked examples: the double-pendulum row is coarse-first rather than the public  $10^{-4}$  policy, and the four-link/slider-crank public-horizon rows are kinematic/reaction residual rows rather than accepted dynamic-order rows. Because a separate residual-to-error transfer theorem remains open, residual-only rows are not promoted to order claims. The external same-test campaign remains incomplete: full source-

policy dynamic-comparison rows remain unaccepted, and the paper makes no external-superiority claim. Equivalently, the source-policy dynamic-order count is nevertheless 0/4 for the external source-paper rows; this count is a progress boundary, not a demotion of the accepted local dynamic-order evidence.

After the coarse common-reference replay was repaired to use fixed public-code times  $t_i = ih$ , the result package also records a bounded common-reference matrix. All 44 common-reference summary rows pass the row-level policy checks, and the global comparison policy passes 14/14 checks, but a stricter all-example consistency check allows zero paper-level external direct-error claims. Thus the replay is a shared coarse final-state diagnostic, not a reproduction of the source papers' default time horizons, output policies, error norms, or  $10^{-4}$  policies.

*Benchmark-policy dictionary.* The bounded common-reference policy used in Tables A.23–A.24 fixes  $T = 0.1$ ,  $h \in \{0.1, 0.05, 0.025\}$ , and  $h_{\text{ref}} = 0.0125$  for every runnable method. The single-pendulum row uses the analytic exact final state and a final  $L^2$  error. The double-pendulum row uses the local Gauss6/FullVA  $h_{\text{ref}} = 0.0125$  final state and a final  $L^\infty$  error. The four-link and slider-crank rows use closed-loop true-dynamic common-reference trajectories at  $h_{\text{ref}} = 0.0125$  and then apply the same order/error arithmetic to all runnable methods. Source-policy rows, by contrast, must follow each source paper or public code on its own horizon, time grid, reference solution, error norm, tolerance, and output variables. This dictionary is the reason the common-reference table can be arithmetically reconciled while source-policy external-superiority rows remain open. Table A.23 summarizes the nearest nonlocal bounded common-reference error evidence, and Table A.24 reports the full runnable method-by-example order/error matrix.

We therefore separate two comparison statements. The finite-grid common-reference diagnostic is assembled for arithmetic coverage of the required matrix: all required method labels are accepted, alias-resolved, or source-backed scope-excluded, and the 44-cell table is recomputed from 132 raw rows with zero summary mismatches. The nonlocal cells remain bounded common-reference diagnostics because they do not reproduce every source paper's default horizon, tolerance, output policy, or error norm. These diagnostic comparisons are not promoted to a CMAME paper-level external-superiority claim. The stronger source-policy statement remains a non-claim, and no common-reference count is promoted to a source-paper

superiority claim.

This paragraph separates local-candidate availability from source-policy closure. For the 2021 absolute-coordinate suite, the public-code path, the `body.r/body.dr/body.ddr` output map, and the endpoint time-grid convention have been extracted and checked; in that bounded sense, source identity is now closed only for the diagnostic row map, not for source-policy closure. The remaining gap is promotion of local Gauss6/FullVA rows under the same source-policy dynamic-order contract. Table A.22 records why the existing local candidates are not promoted: the single-pendulum public-grid tranche is floor limited, the double-pendulum row is a coarse self-reference run, and the four-link/slider-crank true-dynamic rows are bounded  $T = 0.1$  diagnostics rather than the public  $T = 3$  source-policy rows. For the 2022 half-implicit suite, the bounded  $T = 0.1$  replay has 24/24 successful rows, while the  $T = 8$  coarse replay has 18/24 successful rows and only four of eight complete form-model groups. For the original TFE pendulum suite, bounded runner and endpoint-sensitivity diagnostics exist, but a source-equivalent DAE/friction/output runner has not completed the original rows. Consequently no local method row is accepted under the external source-policy dynamic-order contract, and no source-policy-closed external same-test error row is established.

Table A.22: RA2021 local-candidate promotion boundary.

| Example         | Local candidate evidence                                                                                                                           | Why not promoted to RA2021 source-policy row                                                                                              |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Single pendulum | Public $T = 3$ row trio at $h = \{10^{-2}, 10^{-3}, 10^{-4}\}$ with $h_{\text{ref}} = 10^{-3}$ ; finest velocity error is $7.01 \times 10^{-15}$ . | The row is roundoff/floor limited; observed order is not a stable source-policy dynamic-superiority certificate.                          |
| Double pendulum | Coarse $T = 3$ self-reference row with $h = \{0.1, 0.05, 0.025\}$ , $h_{\text{ref}} = 0.0125$ , and position/velocity order 7.951/7.042.           | The step grid and reference do not match the RA2021 double-pendulum policy $h = \{0.01, 0.002, 0.001\}$ with $h_{\text{ref}} = 10^{-4}$ . |
| Four-link       | True-dynamic local row at $T = 0.1$ , $h = \{0.1, 0.05, 0.025\}$ , with position/velocity order 5.955/6.085.                                       | The public $T = 3$ rows are kinematic/reaction checks; the true-dynamic row is not the RA2021 source-policy horizon.                      |
| Slider-crank    | True-dynamic local row at $T = 0.1$ , $h = \{0.1, 0.05, 0.025\}$ , with position/velocity order 6.164/7.341.                                       | The public $T = 3$ rows are kinematic/reaction checks; the true-dynamic row is not the RA2021 source-policy horizon.                      |

The full common-reference table is arithmetic-consistent but not, by itself, a source-policy reproduction. An independent recomputation from the 132

Table A.23: Bounded common-reference velocity evidence from the current result pack; diagnostic only, not external-superiority evidence.

| Example         | Local velocity order | Local finest velocity error | Nearest nonlocal method    | Nearest nonlocal finest error |
|-----------------|----------------------|-----------------------------|----------------------------|-------------------------------|
| Single pendulum | 5.801                | $1.155 \times 10^{-12}$     | velocity partitioning $rA$ | $3.163 \times 10^{-11}$       |
| Double pendulum | 6.010                | $2.346 \times 10^{-13}$     | 2021 public $rA$           | $4.924 \times 10^{-5}$        |
| Four-link       | 6.085                | $1.093 \times 10^{-11}$     | velocity partitioning $rA$ | $1.087 \times 10^{-8}$        |
| Slider-crank    | 7.341                | $9.189 \times 10^{-12}$     | velocity partitioning $rA$ | $1.332 \times 10^{-7}$        |

Table A.24: All runnable common-reference velocity rows. Each entry is observed velocity order / finest-step velocity error on the fixed-grid  $h = \{0.1, 0.05, 0.025\}$  sweep with reference  $h = 0.0125$ ; the table is a diagnostic common-reference matrix, not a source-policy superiority table.

| Method                               | Single pendulum                 | Double pendulum                 | Four-link                       | Slider-crank                    |
|--------------------------------------|---------------------------------|---------------------------------|---------------------------------|---------------------------------|
| Gauss6/FullVA                        | 5.801 / $1.155 \times 10^{-12}$ | 6.010 / $2.346 \times 10^{-13}$ | 6.085 / $1.093 \times 10^{-11}$ | 7.341 / $9.189 \times 10^{-12}$ |
| HI 2022 $rA$                         | 1.005 / $7.725 \times 10^{-2}$  | 1.049 / $4.944 \times 10^{-5}$  | 1.015 / $1.970 \times 10^{-1}$  | 0.941 / $3.281 \times 10^{-2}$  |
| HI 2022 $rA_{\text{half}}$           | 1.841 / $7.725 \times 10^{-2}$  | -0.799 / 2.970                  | 1.590 / $1.970 \times 10^{-1}$  | 2.459 / $3.281 \times 10^{-2}$  |
| 2021 public $rA$                     | 0.617 / $1.305 \times 10^1$     | 1.052 / $4.924 \times 10^{-5}$  | 1.015 / $1.970 \times 10^{-1}$  | 0.941 / $3.281 \times 10^{-2}$  |
| 2021 public $re$                     | 0.617 / $1.305 \times 10^1$     | 1.052 / $4.924 \times 10^{-5}$  | 2.894 / $1.458 \times 10^{-2}$  | 0.940 / $3.281 \times 10^{-2}$  |
| 2021 public $rp$                     | 0.517 / $1.499 \times 10^1$     | 0.361 / $5.241 \times 10^{-5}$  | 2.894 / $1.458 \times 10^{-2}$  | 0.940 / $3.281 \times 10^{-2}$  |
| TFE Newmark- $\beta$                 | 0.901 / $1.479 \times 10^1$     | 1.787 / $4.246 \times 10^{-4}$  | 1.872 / $1.672 \times 10^{-2}$  | 2.051 / $1.531 \times 10^{-3}$  |
| TFE $m = 1$                          | 1.021 / $1.498 \times 10^1$     | 1.996 / $2.708 \times 10^{-4}$  | 3.866 / $5.253 \times 10^{-4}$  | 2.481 / $6.264 \times 10^{-4}$  |
| TFE $m = 2$                          | 0.054 / $1.102 \times 10^2$     | 1.870 / $8.711 \times 10^{-5}$  | 1.021 / $6.711 \times 10^{-4}$  | 2.290 / $9.881 \times 10^{-5}$  |
| TFE trapezoidal                      | 1.024 / $1.498 \times 10^1$     | 2.000 / $2.708 \times 10^{-4}$  | 3.888 / $5.306 \times 10^{-4}$  | 2.329 / $7.328 \times 10^{-4}$  |
| VP 2024 coordinate partitioning $rA$ | 0.000 / $3.163 \times 10^{-11}$ | 1.000 / $2.165 \times 10^{-3}$  | -0.001 / $1.087 \times 10^{-8}$ | -0.000 / $1.332 \times 10^{-7}$ |

raw common-reference rows reproduces all 44 summary order/error entries with zero summary mismatches. The remaining concern is interpretive rather than arithmetical: 15 nonlocal method/example rows are flagged for source-policy review, and 10 of those rows have small position error but large velocity error. This pattern is consistent with a velocity-output, state-map, time-grid, or reference-policy mismatch and therefore cannot support an external-superiority claim. Table A.25 records the resulting claim boundary.

The flagged-row count should not be read as the complete set of open external claim rows. All 40 nonlocal method/example rows remain bounded common-reference diagnostics rather than source-policy-closed reproductions; the 15 flagged rows are the anomaly or high-risk subset that requires special attention before any separate external-superiority statement is considered. This distinction keeps the order/error comparison and the source-policy claim separate throughout the paper.

The all-example source-policy check expands this diagnosis from category counts to the actual method/example coverage. It checks all 15 flagged summary rows and their 45 raw coarse rows at  $h = \{0.1, 0.05, 0.025\}$ . The flagged rows cover all four examples: 8 single-pendulum rows, 2 double-

Table A.25: Source-policy diagnosis for flagged common-reference baseline rows.

| Diagnostic category                                 | Rows | Affected examples                                         | Claim-boundary interpretation                                                                           |
|-----------------------------------------------------|------|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Position aligned, velocity mismatched               | 10   | single pendulum, four-link                                | Likely velocity-output, state-map, or reference-policy mismatch; not a method-failure claim.            |
| Single-pendulum velocity-output policy suspect      | 7    | single pendulum                                           | Original/Taves-Kissel-Negrut/TFE pendulum rows require source setup and velocity-output policy closure. |
| Near reference floor; order not identifiable        | 3    | single pendulum, four-link, slider-crank                  | Errors are too close to the shared reference floor for a meaningful observed order statement.           |
| Low order but small error; wider step window needed | 4    | single pendulum, double pendulum, four-link, slider-crank | The row is diagnostic only until a wider or source-policy step window is reported.                      |
| Negative order with large error                     | 1    | double pendulum                                           | The fixed-grid replay or selected form requires source-policy recheck before comparison.                |

pendulum rows, 4 four-link rows, and 1 slider-crank row. Table A.26 records the suite-level closure boundary. A stricter all-method/all-example consistency check then checks the complete 44-cell common-reference table and its 132 raw rows; it keeps all 44 cells as bounded diagnostics but allows zero strict external error-claim rows. In particular, the common-reference replay is valid for the finite-grid diagnostic, but no suite has yet closed the stronger source-policy reproduction required for an external-superiority claim.

For the TFE source evidence, we separate the friction evidence into two levels. The source text identifies the Brown–McPhee velocity-based continuous friction model, reports  $\mu_s = 0.3$  and  $\mu_d = 0.2$ , and uses  $h_{\text{ref}} = 10^{-4}$  for source reference solutions; its detailed continuous-friction law is delegated to the source-paper friction references [1, 28]. The model check therefore treats the encoded Brown–McPhee torque law as a candidate published formula structure, not as a source-code-equivalent friction implementation. The transition-velocity, normal-load coupling, friction Jacobian, and DAE multiplier policies remain source-policy boundaries.

Source-policy boundary summary:

Brown–McPhee velocity-based continuous friction model;

candidate published formula, not source-code-equivalent friction implemen-

Table A.26: All-example source-policy status for the flagged baseline rows.

| Suite                                       | Rows | Common-reference row source                                                                                                               | Closure boundary                                                                                                                                                                          |
|---------------------------------------------|------|-------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2021<br>absolute-coordinate<br>public suite | 5    | Public setup and $rA/rp/r\epsilon$ steppers, fixed-grid replay, and <code>body.r/body.dr/body.ddr</code> output fields.                   | Source identity is checked; promotion still requires local Gauss6/FullVA source-policy dynamic rows, norm/output binding, runtime/Newton binding, and an independent verification record. |
| Original TFE<br>pendulum formulas           | 4    | Surrogate common-reference row only: RA2021 public setup with locally installed Newmark and TFE formula proxies, not the source pendulum. | Encode the original pendulum setup, source error norm, Brown–McPhee friction law, output variables, and horizon, or keep these rows demoted.                                              |
| HI 2022 half-implicit<br>suite              | 3    | Public setup and half-implicit steppers under bounded fixed-grid replay.                                                                  | Run the full $T = 8$ public policy or explicitly demote the bounded pilot rows.                                                                                                           |
| VP 2024 coordinate<br>partitioning          | 3    | Coordinate-partitioning proxy on the RA2021 public setup.                                                                                 | Resolve the independent velocity-partitioning code path or keep the row as proxy/common-reference evidence only.                                                                          |

tation;  
 source-policy boundaries;  
 Brown–McPhee formula is not fixed;  
 source-code-equivalent Brown–McPhee transition law;  
 transition velocity  $v_t$ ;  
 velocity scale in the  $v/v_t$  Stribeck factor;  
 not the normal-load/multiplier coupling;  
 implementation choice, not source-policy evidence;  
 candidate-friction DAE trajectory contract records 12 method-grid rows and  
 56 step-residual rows with nonpositive friction power;  
 do not implement a monolithic source-policy DAE runner;  
 source-policy runner acceptance contract has six simultaneous parts;  
 transition law, including transition velocity;  
 velocity-based continuous friction model;  
 row-promotion rules binding error, order, runtime, and work metrics;  
 monolithic absolute-coordinate DAE time integrator;  
 not an accepted source-policy reproduction.

More concretely, the Brown–McPhee formula is not fixed by  $\mu_s$ ,  $\mu_d$ , and  $R$  alone. It also depends on a transition velocity  $v_t$ , the velocity scale in the  $v/v_t$  Stribeck factor, and on how the tangential velocity and normal reaction are supplied by the constrained DAE. The source-pendulum text records  $\mu_s$ ,  $\mu_d$ ,  $R$ , and  $h_{\text{ref}}$ , but not  $v_t$ , not the normal-load/multiplier coupling, and not the friction-Jacobian convention. Choosing the local surrogate’s Stribeck velocities 0.05, 0.5, or 1.0 would therefore be an implementation choice, not source-policy evidence. A bounded transition-velocity sensitivity check makes this boundary numerical rather than merely documentary: over Stribeck velocities 0.05, 0.5, and 1.0, with 0.5 as the baseline, the  $T = 0.024$ ,  $h_{\text{ref}} = 10^{-4}$  candidate DAE check changes the endpoint by at most  $2.90 \times 10^{-6}$  in coordinate and  $2.36 \times 10^{-4}$  in velocity. The check records three sensitivity rows, 36 contract rows, and zero source-policy rows, so the missing transition velocity is treated as a material unresolved source-law parameter rather than an innocuous transcription detail. The positive evidence from the surrogate-friction DAE layer is narrower: the candidate-friction DAE trajectory contract records 12 method-grid rows and 56 step-residual rows, all finite, with candidate DAE residuals below  $10^{-9}$  and nonpositive friction power. These checks certify local dissipativity and absolute-coordinate residual consistency for the encoded candidate law only. They do not certify a

source-code-equivalent Brown–McPhee law, do not implement a monolithic source-policy DAE runner, and do not close any source-policy row.

Consequently, the source-policy runner acceptance contract has six simultaneous parts. A promoted TFE row would need a source-code-equivalent Brown–McPhee transition law, including transition velocity, normal-load coupling, friction Jacobian, and multiplier policy; a monolithic absolute-coordinate DAE time integrator rather than a stepwise residual replay; source grid, output norm, and endpoint/output-sampling binding for  $T = 10$  and  $h_{\text{ref}} = 10^{-4}$ ; nonlinear solve, Newton, and tolerance-policy binding; method-runner equivalence for the TFE  $m = 1, 2, 3$ , Newmark- $\beta$ , trapezoidal, and local Gauss6/FullVA comparator rows; and row-promotion rules binding error, order, runtime, and work metrics. Equivalently, the present package leaves two documentary blocks open without requiring a new long numerical campaign—the source-code-equivalent Brown–McPhee law and the full- $T = 10$  endpoint/output-sampling policy for endpoint-incompatible step sizes—and four execution-level blocks open: a monolithic source-policy DAE integrator, source-policy method runners for the TFE and one-step comparator rows, a Gauss6/FullVA source-pendulum DAE comparator, and accepted work/precision row binding. Because these blocks are not jointly present, the candidate DAE trajectory evidence remains a diagnostic record of what is missing, not an accepted source-policy reproduction row.

The source check also includes the Appendix-B coefficient formulas for TFE  $m = 1$ ,  $m = 2$ , and  $m = 3$  Gauss–Lobatto against the source equations (40), (41), and (43). The recorded coefficient certificate has three checked rows and zero maximum absolute difference for the extracted source parameter choices. This closes coefficient-transcription evidence only; it does not close source-policy method-runner equivalence or any source-policy reproduction row. Here, Appendix-B coefficient formulas denotes this coefficient-transcription certificate only, not a closed source-policy method runner. The diagnostic chain also includes a full- $T = 10$  coarse formula probe for the Appendix-B  $m = 3$  Gauss–Lobatto target. That probe is finite and residual-converged for one checked row and records the formal fifth-order target, but it remains a non-source-policy formula diagnostic because it does not invoke the source  $h = 10^{-4}$  reference campaign. Restated as a source-policy boundary, the full- $T = 10$  coarse formula probe targets the Appendix-B  $m = 3$  Gauss–Lobatto target, records a formal fifth-order target, and does not invoke the source  $h = 10^{-4}$  reference campaign. The source-

reference side is therefore checked separately. A full- $T = 10$  frictionless source-reference feasibility probe integrates the extracted source pendulum with  $h_{\text{ref}} = 10^{-4}$  and compares it against a  $h = 5 \times 10^{-5}$  check run, using 100000 and 200000 RK4 steps, respectively. The  $h_{\text{ref}} = 10^{-4}$  final state is  $(\theta, \omega) = (-2.8790156956337563, -1.2836123173308356)$ , and the check-run mismatches are  $3.82 \times 10^{-14}$  in coordinate and  $2.49 \times 10^{-13}$  in velocity. This closes only source-reference feasibility for the original pendulum time horizon. It is not source-policy method-runner equivalence for Newmark, trapezoidal, or TFE, it does not cover the Brown–McPhee friction row, and it does not authorize external-superiority claims; no source-policy row is promoted.

The same source-pendulum candidate route now includes a Gauss6/FullVA row on the extracted source output policy. On the frictionless  $T = 1$  run with  $h_{\text{ref}} = 2.5 \times 10^{-4}$  and  $h = \{0.1, 0.05, 0.025\}$ , the coordinate, velocity, and Frobenius pairwise orders are (5.98, 6.10), (6.02, 6.03), and (5.98, 6.10), respectively, with maximum Newton residual below  $10^{-12}$ . This is same-test candidate evidence for the Gauss collocation order target on the source pendulum output policy. The candidate frictional row on the same grid shows reduced pairwise orders (about 1.6–2.3), so it is recorded only as a friction-feature diagnostic. Neither row is an absolute-coordinate FullVA DAE source-policy reproduction, and the TFE source-policy closure count remains zero. A separate same-test work/precision diagnostic runs Newmark- $\beta$ , trapezoidal, TFE  $m = 1, 2, 3$ , and Gauss6/FullVA on this same frictionless source-pendulum  $T = 1$  grid. It records 18 raw rows and six method summaries with runtime and Newton-iteration work proxies; the diagnostic velocity-order floors are about 6.02 for the Gauss6/FullVA candidate and 5.19 for the TFE  $m = 3$  Gauss–Lobatto formula target. This diagnostic improves the fair same-grid comparison, but it remains candidate-level evidence: the source-policy method-runner equivalence flag and the external-superiority claim flag are both false.

For the original TFE pendulum suite, the source-model evidence also includes a bounded active-row candidate smoke test. It uses the extracted pendulum parameters and source output/error policy with  $h_{\text{ref}} = 10^{-4}$ , but only on the short frictionless interval  $T = 0.024$  and the source-shaped grid  $h = \{0.012, 0.006, 0.003\}$ . The row is therefore useful for checking whether the installed Newmark, trapezoidal, and TFE formula runners have coherent local order/error behavior, but it is not the full  $T = 10$  original-source campaign and it closes zero source-policy rows. Table A.27 records this bounded diagnostic.

Table A.27: Bounded active-row candidate smoke for the original TFE pendulum rows. The run uses  $T = 0.024$ ,  $h_{\text{ref}} = 10^{-4}$ , and  $h = \{0.012, 0.006, 0.003\}$ ; it is not a full  $T = 10$  source-policy reproduction.

| Original-row method | Formula target | Velocity pair orders | Coordinate pair orders | Finest velocity error   | Finest coordinate error | Claim boundary               |
|---------------------|----------------|----------------------|------------------------|-------------------------|-------------------------|------------------------------|
| Newmark- $\beta$    | 2              | 1.973, 1.997         | 2.060, 2.022           | $1.650 \times 10^{-10}$ | $2.584 \times 10^{-12}$ | Bounded candidate smoke only |
| TFE $m = 1$         | 1              | 1.934, 1.915         | 1.000, 1.000           | $1.856 \times 10^{-10}$ | $5.742 \times 10^{-7}$  | Bounded candidate smoke only |
| TFE $m = 2$         | 3              | 4.000, 3.998         | 3.749, 3.609           | $6.482 \times 10^{-14}$ | $4.441 \times 10^{-15}$ | Bounded candidate smoke only |
| Trapezoidal         | 2              | 1.973, 1.997         | 2.078, 2.028           | $1.650 \times 10^{-10}$ | $1.987 \times 10^{-12}$ | Bounded candidate smoke only |

The same source-model evidence record also includes a non-heavy full-horizon candidate probe at the original TFE pendulum horizon  $T = 10$ . This diagnostic uses  $h = \{0.1, 0.05, 0.025\}$  and a coarse reference  $h_{\text{ref}} = 0.0125$ , so it tests full-horizon large-step behavior without invoking the extracted  $h_{\text{ref}} = 10^{-4}$  source-policy reference. All four active rows are finite and Newton-residual stable, but the table remains a candidate-runner diagnostic rather than a source-policy reproduction; it closes zero source-policy rows.

Table A.28: Full-horizon coarse candidate probe for the original TFE pendulum rows. The run uses  $T = 10$ ,  $h_{\text{ref}} = 0.0125$ , and  $h = \{0.1, 0.05, 0.025\}$ ; it does not invoke the source-policy  $h_{\text{ref}} = 10^{-4}$  reference.

| Original-row method | Formula target | Velocity pair orders | Coordinate pair orders | Finest velocity error  | Finest coordinate error | Claim boundary                  |
|---------------------|----------------|----------------------|------------------------|------------------------|-------------------------|---------------------------------|
| Newmark- $\beta$    | 2              | 1.991, 1.998         | 2.028, 2.007           | $4.642 \times 10^{-3}$ | $1.956 \times 10^{-3}$  | Full- $T$ coarse candidate only |
| TFE $m = 1$         | 1              | 1.276, -0.077        | 2.623, 5.795           | $6.801 \times 10^{-3}$ | $5.226 \times 10^{-5}$  | Full- $T$ coarse candidate only |
| TFE $m = 2$         | 3              | 2.927, 2.952         | 2.694, 2.824           | $2.375 \times 10^{-6}$ | $3.346 \times 10^{-7}$  | Full- $T$ coarse candidate only |
| Trapezoidal         | 2              | 1.993, 1.998         | 2.021, 2.005           | $3.558 \times 10^{-3}$ | $1.504 \times 10^{-3}$  | Full- $T$ coarse candidate only |

## Appendix B. Reproducibility package

The flat LaTeX archive contains the editable Elsevier source, the compiled manuscript, the thirteen figures, and declaration files. The companion repository-facing reproducibility package contains the scripts needed to check

Table A.29: Cross-paper evidence accumulated without treating it as an external superiority result.

| Tranche                                               | Rows                                 | Main metric                                                                                                                                                                                                                                                                                                                                        | Interpretation                                                                                                   |
|-------------------------------------------------------|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| 2021 public $rA/rp/r\epsilon$ baselines               | 27 + 9 order rows;<br>12 timing rows | Public $T = 3$ policies;<br>$h = 10^{-2}, 10^{-3}, 10^{-4}$ except the<br>double-pendulum dynamic<br>self-reference sweep<br>$h = 10^{-2}, 2 \times 10^{-3}, 10^{-3}$ .<br>$T = 3, h = 10^{-2}, 10^{-3}, 10^{-4}$ ; finest<br>position/velocity errors<br>$2.584 \times 10^{-14} / 7.012 \times 10^{-15}$ .                                        | Baseline<br>execution<br>evidence only.                                                                          |
| Gauss6/FullVA single public<br>horizon                | 3 rows                               |                                                                                                                                                                                                                                                                                                                                                    | Public-policy<br>rows closed, but<br>order is<br>reference-floor<br>limited.                                     |
| Gauss6/FullVA double public<br>horizon, coarse        | 3 rows                               | $T = 3, h = 0.1, 0.05, 0.025$ ;<br>reference $h = 0.0125$ ;<br>position/velocity orders<br>$7.951 / 7.042$ .                                                                                                                                                                                                                                       | Coarse-first<br>order/work<br>evidence, not the<br>public $10^{-4}$<br>policy.                                   |
| Public double coarse<br>same-window baselines         | 9 rows, 6 ok                         | $rA/r\epsilon$ at the same $T = 3$ ,<br>$h = 0.1, 0.05, 0.025$ , reference<br>$h = 0.0125$ give position/velocity<br>orders $0.703 / 0.754$ ; $rp$ fails Newton<br>at all three coarse steps.<br>$T = 3, h = 0.1, 0.05, 0.025$ ,<br>reference $h = 0.0125$ ;<br>Gauss6/FullVA position order<br>$6.054$ ; public velocity orders about<br>$1.21$ . | Large-step<br>baseline evidence;<br>not the source<br>$10^{-4}$ policy.                                          |
| Gauss6/FullVA and public<br>single coarse same-window | 12 raw rows; 4<br>summary rows       |                                                                                                                                                                                                                                                                                                                                                    | Coarse-first<br>evidence; no<br>default $10^{-4}$ run<br>or superiority<br>claim.                                |
| Gauss6/FullVA closed-loop<br>public mechanisms        | 12 rows                              | Selected and public-horizon residual<br>trios for four-link/slider-crank;<br>finest public residuals<br>$5.136 \times 10^{-13} / 1.113 \times 10^{-14}$ .                                                                                                                                                                                          | Residual<br>evidence, not<br>dynamic<br>order/work<br>superiority.                                               |
| Closed-loop surrogate and floor<br>analysis           | 2+2 rows                             | Selected-window residual-to-error<br>ratios for four-link/slider-crank; no<br>source-policy dynamic-order row is<br>accepted; the floor analysis has two<br>velocity/acceleration evidence rows<br>and two position/reference-floor<br>limitations.                                                                                                | Narrows the<br>boundary; not<br>external<br>superiority.                                                         |
| 2022 half-implicit pilot                              | 24 rows                              | Bounded $T = 0.1$ ,<br>$h = 0.02, 0.01, 0.005$ for $rA/rA_{\text{half}}$<br>on four examples.                                                                                                                                                                                                                                                      | Source-family<br>candidate; full<br>$T = 8$ policy<br>remains open.                                              |
| Closed-loop coarse trajectory<br>diagnostics          | 6 rows                               | Four-link primary-state orders<br>$5.955 / 5.955 / 6.085 / 5.971$ ;<br>slider-crank orders<br>$6.164 / 6.159 / 7.341 / 6.426$ at<br>$T = 0.1, h = 0.1, 0.05, 0.025$ ,<br>reference $h = 0.0125$ .                                                                                                                                                  | Local coarse<br>trajectory<br>evidence only; not<br>accepted<br>dynamic-order<br>proof or public<br>superiority. |
| Bounded common-reference<br>work/precision diagnostic | 24 rows                              | Local Gauss6/FullVA and public<br>$rA/rp/r\epsilon$ rows use the same finite<br>accepted-method endpoint<br>reference; the fixed-reference<br>diagnostic gap count is zero.                                                                                                                                                                        | Figure 8; no<br>external<br>superiority claim.                                                                   |
| Coarse-first readiness diagnostic                     | 4 rows                               | Single and double pendulum are<br>coarse same-window order/time<br>ready; four-link and slider-crank<br>have local coarse trajectory<br>diagnostics and strict<br>common-reference work/precision<br>rows; the $10^{-4}$ policy is outside the<br>default evidence package.                                                                        | Readiness<br>diagnostic, not a<br>superiority claim.                                                             |
| Coverage matrix                                       | 48 rows                              | 32 complete, 0 partial; no<br>external-superiority claim.                                                                                                                                                                                                                                                                                          | Diagnostic<br>coverage table,<br>not a paper claim.                                                              |

the reported tables. The paper-level checks read existing result records unless a full numerical rebuild is explicitly requested; this keeps editorial checks separate from computational campaigns. Companion records encode the accepted claim scope and the non-claims, but the mathematical and numerical arguments in the paper are the authority.

The current reproducibility package is therefore two-tiered. The reviewer-facing minimal candidate is a 10-file package with one 158-line Python replay core; it is a shrink target for the reported-result records and proof-dependency records, not a complete runner-centered source-policy archive. The local accepted-row companion is human-runnable and self-contained for the four local examples and the 12 accepted local rows. A broader external source-policy runner archive would require the 40 source-policy rows, the TFE source-policy runner, and regeneration of both the accepted local rows and any promoted source-policy rows from a standalone runner. That archive is outside the current paper package.

Table B.30: External reproduction and package boundaries retained by the paper package.

| Boundary                        | Current evidence state                                                                                                                                                                                                              | Manuscript consequence                                                                                                                            |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| External same-test reproduction | External source-policy reproduction has not been established on the source paper tests, and no external rows are treated as source-policy closed.                                                                                   | No source-paper superiority or source-policy direct-error claim is made.                                                                          |
| Original TFE runner package     | A complete source-policy TFE runner package would need the source pendulum DAE runner, TFE/comparator source-policy method runners, Gauss6 source-policy runner, full-T10 endpoint policy, and accepted work-precision source rows. | TFE rows are formula/common-reference diagnostics, not full-TFE or source-policy runner evidence.                                                 |
| Runner archive boundary         | The local accepted-row runner package covers the bounded replay/provenance role for the accepted rows; a full source-policy runner archive is outside this package.                                                                 | The package supports accepted-row provenance while keeping full source-policy runner evidence outside the accepted theorem and comparison claims. |

External reproduction/package boundary: the three rows in Table B.30 are scope boundaries, not theorem hypotheses. They do not change the conditional theorem proof; they keep source-policy reproduction, full-TFE runner evidence, and a full source-policy runner archive outside the accepted theorem and comparison claims. Notation boundary: theorem P6 denotes only the compact-tube solver-scale interface  $\eta_h^{\text{tube}} \leq c_\eta h^7$ , whereas the separate original-TFE source-policy DAE-runner/package gap is an external re-

production boundary, not a theorem condition.

The implementation record identifies `residual_cylindrical_chain`, `R_VALUE`, and `R_JAC` as the implemented FullVA residual and Jacobian path. This supports independent reruns of the reported evidence; it is not a substitute for the proof.

#### **Comparison reconciliation statement.**

The finite-grid common-reference diagnostic is arithmetically complete, but the nonlocal cells remain diagnostic comparison records. These rows remain diagnostic scope records and are not promoted to a CAME paper-level external-superiority claim.

*Lean development.* The source package contains the Lean 4 development (directory `lean/`, Mathlib pinned by `lake-manifest.json`) that machine-checks the algebraic and perturbation lemmas of Section 7. It builds with `lakebuild` after `lakeexecacheget`; the command file `scripts/Axioms.lean` prints, for every theorem in Table B.35, that only the standard axioms `propext`, `Classical.choice`, and `Quot.sound` are used and that no proof is left open. The exact-stage-identity check in the package records the result of that command when the toolchain is present.

The source package includes the command file for these read-only checks.

### **Appendix C. Supplementary source package**

The Elsevier source package contains the editable `elsarticle` manuscript source, the compiled manuscript PDF, the separate highlights file, the declaration statements, and the thirteen figure files used in this paper. A supporting order-scope record separates accepted dynamic order from four-example mechanism coverage. A flat Editorial-Manager-oriented source folder, `cmame_submission_flat`, is also supplied. In that folder, the submission TeX source and all referenced figure files live at the same directory level, avoiding a dependency on the repository’s figure subdirectory. The reproducibility package is therefore two-tiered: a compact 10-file package with one 158-line Python replay core records the bounded local replay, while the full development repository remains provenance for the broader development record. The local accepted-row companion is human-runnable and records 12 accepted local rows, but it is not a complete runner-centered source-policy archive. A full source-policy runner archive is not included; it would require closing the 40 source-policy rows

Table B.31: Algorithm-to-source map for the accepted Gauss6/FullVA execution path.

| Algorithm block                    | Source location                                                                             | Reproduction role                                                                                                           |
|------------------------------------|---------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Mechanism setup and examples       | <code>v047_cylindrical_chain_pipeline/run_v047.py</code>                                    | Builds the cylindrical-chain data and the four ASME-style example rows used by the accepted method-side evidence.           |
| Residual and Jacobian path         | <code>residual_cylindrical_chain, R_VALUE, R_JAC</code>                                     | Defines the 132-row FullVA residual and the single accepted residual/Jacobian path used by Newton.                          |
| One-step solve and trajectory loop | <code>gauss_step, integrate, run_case</code>                                                | Implements the Algorithm 1 stage solve, endpoint reconstruction, endpoint closure, and step diagnostics.                    |
| Method result records              | <code>v047_cylindrical_chain_pipeline/results</code>                                        | Record the h-sweeps, endpoint closure diagnostics, four-example rows, and accepted method-side order/coverage evidence.     |
| Cross-paper comparison records     | <code>v048_cross_paper_same_test_benchmarks/results</code>                                  | Store common-reference diagnostics and external-row status; these records do not authorize source-paper superiority claims. |
| Package-level checks               | <code>validate_paper_package.py--latex</code> and <code>validate_pipeline_outputs.py</code> | Rebuild and check the manuscript package and repository inventory without invoking a new full numerical campaign.           |

Table B.32: Compact reproducibility contract used by the bounded paper package.

| Field                               | Paper-level value                                                                                                                                                                                                                             |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Accepted method                     | Gauss6/FullVA.                                                                                                                                                                                                                                |
| Conditional method order            | Sixth-order path of Theorem 1.                                                                                                                                                                                                                |
| Smooth observed orders              | Position/velocity 7.161/7.066.                                                                                                                                                                                                                |
| Local method-side coverage examples | single pendulum, double pendulum, four-link, and slider-crank.                                                                                                                                                                                |
| Accepted dynamic order rows         | Single and double pendulum.                                                                                                                                                                                                                   |
| Coverage-only for dynamic order     | Four-link and slider-crank; their closed-loop kinematic/reaction rows are not accepted external dynamic-order rows.                                                                                                                           |
| Comparator                          | Local paper-style $m = 3$ Gauss–Lobatto TFE formula target.                                                                                                                                                                                   |
| Comparator expected order           | 5.                                                                                                                                                                                                                                            |
| Numerical comparison scope          | Forty-four method/example common-reference order-error cells, with external rows kept outside source-paper superiority.                                                                                                                       |
| Runner-package status               | The local accepted-row companion is executable for the 12 accepted local rows; a full source-policy runner archive is outside the current package and would require separate source-policy closure and the original-TFE source-policy runner. |
| Non-claims                          | No full TFE replacement, no source-paper superiority, and no paper-level direct-error superiority against external methods.                                                                                                                   |

Table B.33: Cross-example and review records retained with the source package.

| Record                            | Role                                                                                                                                                                                             |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| All-example result matrix         | Checks every one of the 44 method/example cells and separates the 15 nonlocal rows that are sensitive to external-row policy.                                                                    |
| External-row table                | Lists the row-level evidence still needed before any external row can support source-paper superiority.                                                                                          |
| Case reconciliation record        | Separates full-campaign not-run markers from the bounded/coarse evidence already present for the RA2021 and HI2022 families.                                                                     |
| Proof and package-boundary record | Keeps proof assumptions, dynamic-row closure requirements, open obligations, all-example numerical sanity, and current package boundary traceable while external reproduction items remain open. |

Table B.34: Source-package reproducibility and consistency checks.

| Check group                 | Role                                                                                                                                                                                      |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source-format check         | Checks the Elsevier source, PDF, highlights, declarations, clean log, and non-claim scope.                                                                                                |
| Archive-completeness check  | Checks the source archive, required figures, clean logs, and the rule that the full numerical generator is not started by editorial checks.                                               |
| Order-scope check           | Checks that single/double pendulum carry the accepted dynamic-order rows, four-link/slider-crank remain coverage-only for dynamic order, and $10^{-4}$ is not a default execution policy. |
| Numerical-matrix check      | Checks the 44-cell numerical matrix against the common-reference and reconciliation records.                                                                                              |
| External-row check          | Checks the separated nonlocal rows, zero claim-ready external rows, and the non-default- $10^{-4}$ execution rule.                                                                        |
| Proof-dependency check      | Checks proof-label dependency links, theorem-scope assumptions, and the exact-stage-identity gate that pins the compacted proof route.                                                    |
| Scope-consistency review    | Checks that the all-example matrix, proof-dependency record, and stated non-claims use the same claim boundary.                                                                           |
| Package check               | Runs the paper, source-archive, source-comparison, proof-matrix, four-example, and accepted-method package checks without starting a new full numerical campaign.                         |
| Repository-provenance check | Checks the broader repository inventory and records that the full numerical generator was not started by the source-package checks.                                                       |

Table B.35: Manuscript statements and the Lean theorems that check them.

| Manuscript statement                                               | Lean theorem                                                                          |
|--------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Lemma 1, non-dynamic rows<br>(both directions)                     | <code>FullVA.Transition.nondynamic_rows_iff</code>                                    |
| Lemma 1, Newton–Euler rows                                         | <code>NewtonEuler.dynamic_rows_vanish</code>                                          |
| Lemma 3, bound Eq. (27)                                            | <code>uniform_inverse_of_perturbation</code>                                          |
| Lemma 2, root uniqueness                                           | <code>root_unique_of_linearization</code>                                             |
| Lemma 4, bound Eq. (29)                                            | <code>gauss6_step_defect</code>                                                       |
| Butcher conditions $B(6)$ ,<br>$C(3)$ , $D(3)$ ; failure of $B(7)$ | <code>Gauss6.butcher_order_six_hypotheses</code> ,<br><code>Gauss6.not_B_seven</code> |
| Lemma 5                                                            | <code>endpoint_closure_exists</code> ,<br><code>endpoint_correction_bound_h7</code>   |
| Lemma 6                                                            | <code>inexact_newton_output_bound</code>                                              |
| Lemma 7, decay Eq. (32)                                            | <code>simplified_newton_residual_decay</code>                                         |
| Lemma 8                                                            | <code>local_to_global</code> , <code>reported_grid_bound</code>                       |
| Theorem 1, three-term chain<br>and grid bounds                     | <code>local_defect_bound</code> ,<br><code>conditional_sixth_order_grid_bound</code>  |

under their source policies. The reviewer-facing runnable evidence for the narrowed current claim is the `cmame_narrowed_repro_bundle` directory. Its `run_narrowed_repro_bundle.py` entry point replays the paper result matrix and launches the four self-contained local accepted-row runners, while recording source-policy external-row closure and the full source-policy runner archive as open boundaries. Thus the local reproducibility bundle supports the present bounded claim boundary, but it is not a source-policy runner archive and does not promote external-superiority rows.

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